

Supplementary Material 1 for: Low-cost niche construction can create conditions for subdivided populations

Supplementary Analyses

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1 Population maintenance?

First, we checked to see whether all model runs maintained a population across all 3500 ticks.

```
all.exp = unique(data$exp)
```

```
end.tick = unique(data$`tick-limit`)
fd = data %>% filter(ticks == end.tick)
finished.exp = unique(fd$exp)
rm(list = "fd")

#finished.data = data %>% filter(exp %in% finished.exp)
unfinished.data = data %>% filter(!(exp %in% finished.exp))
length(unique(unfinished.data$exp))
```

```
## [1] 0
```

None of our modeled scenarios resulted in population extinction.

```
data = data %>% filter(exp %in% finished.exp)
```

2 Population during model run

2.1 Population density over time

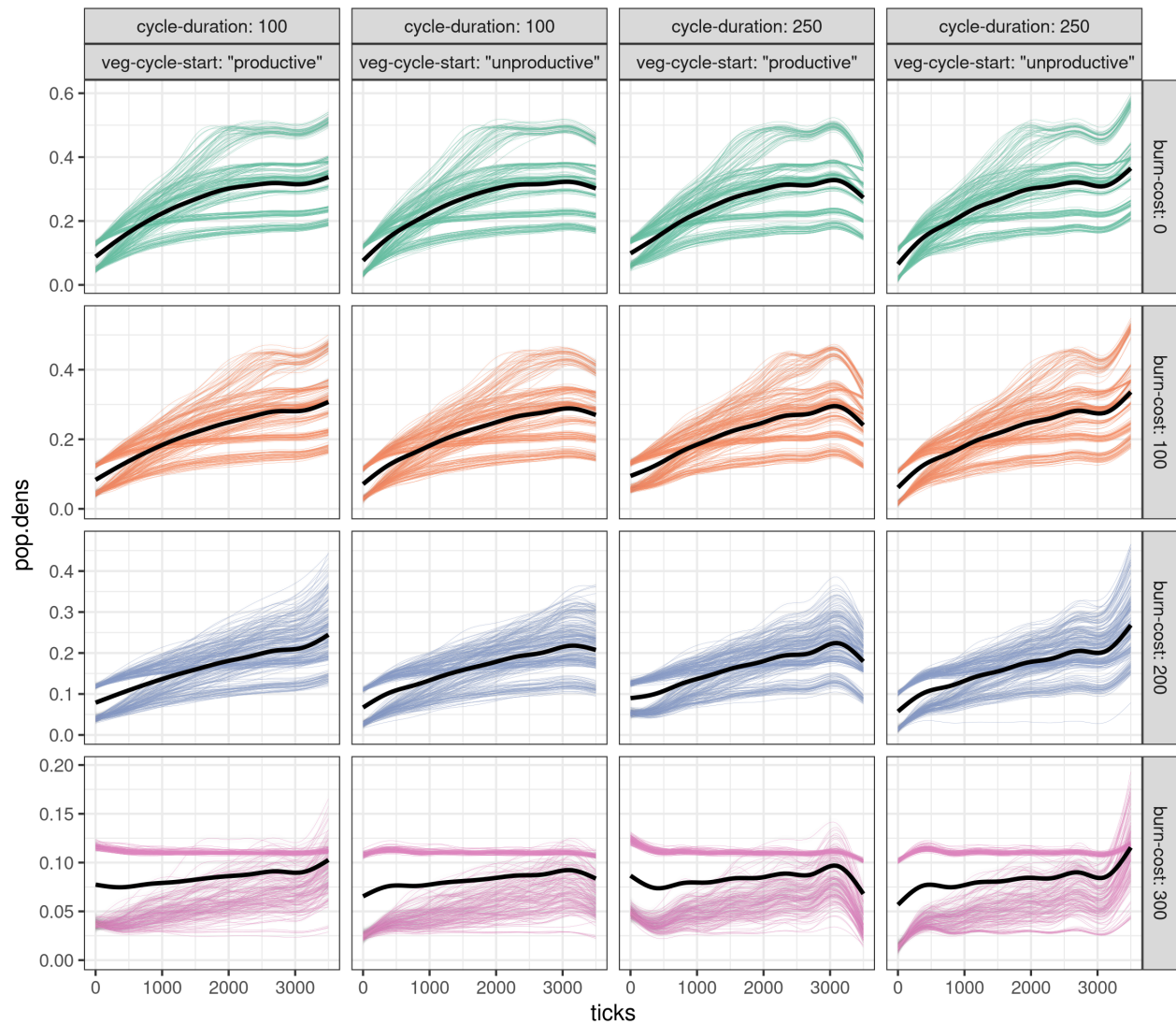


Figure S1: Population density over time

As we can see in Figure S1, population increases over time. There are fluctuations in population based on the climatic condition of the model, with steep drops in population when the landscape is unproductive. However, the overall trend of population increase is evident across all modeled scenarios.

We looked at linear rates of changes within each climate period of the model to get a more detailed understanding of population change during model run.

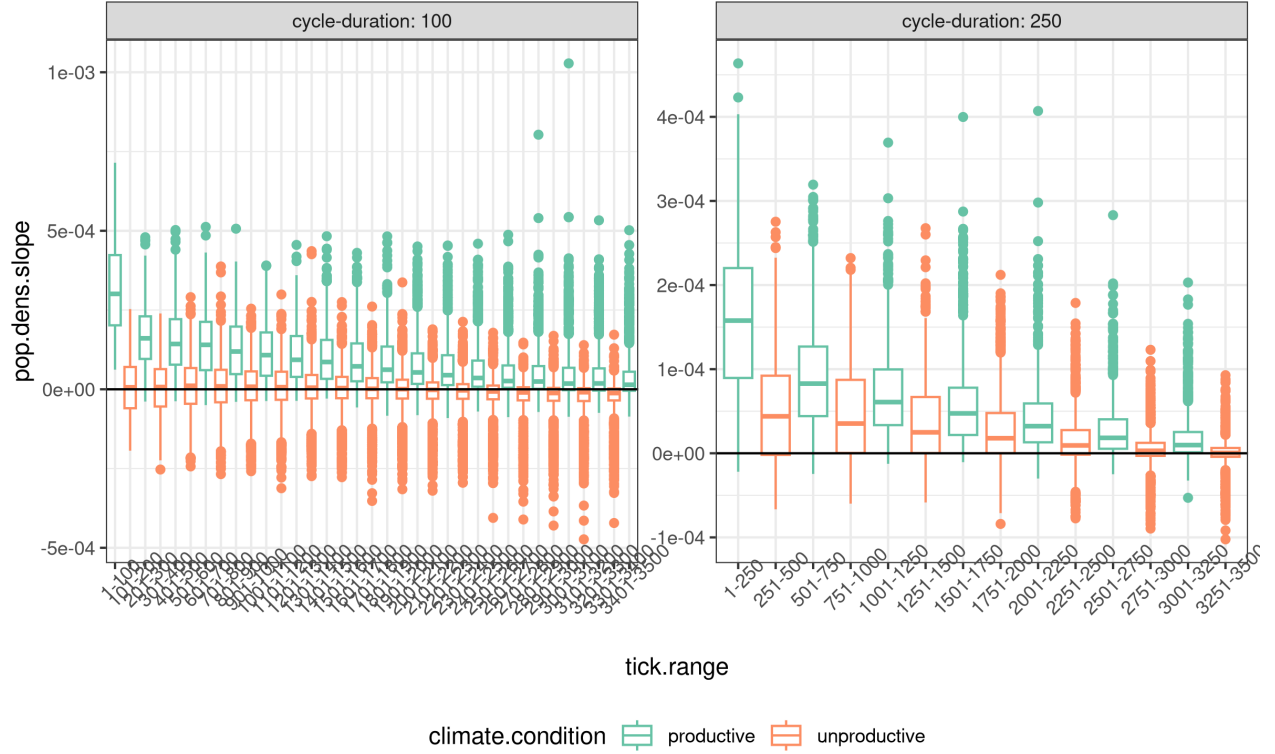


Figure S2: Linear rate of population change within climatic intervals for model runs starting in a productive climatic interval

We find that with short climate cycles, unproductive periods see essentially no change in population, with slopes being essentially zero (Figure S2). With longer climatic cycles, there is slight positive population change during unproductive periods at the beginning of model run.

Conversely, there is pretty consistent population growth during productive periods of the model until populations reach an equilibrium around 3000 ticks. The results show that population grows more quickly during productive periods at the beginning of model run, with rates of change decreasing toward zero by the end of the model, when population reaches equilibrium. This demonstrates the effect of a carrying capacity in the model.

2.2 Impact of amount of forageable land on population density

We can investigate this carrying capacity by looking at how many patches on the landscape are available to be foraged during a given time interval and how this is related to the change in population density over that same interval.

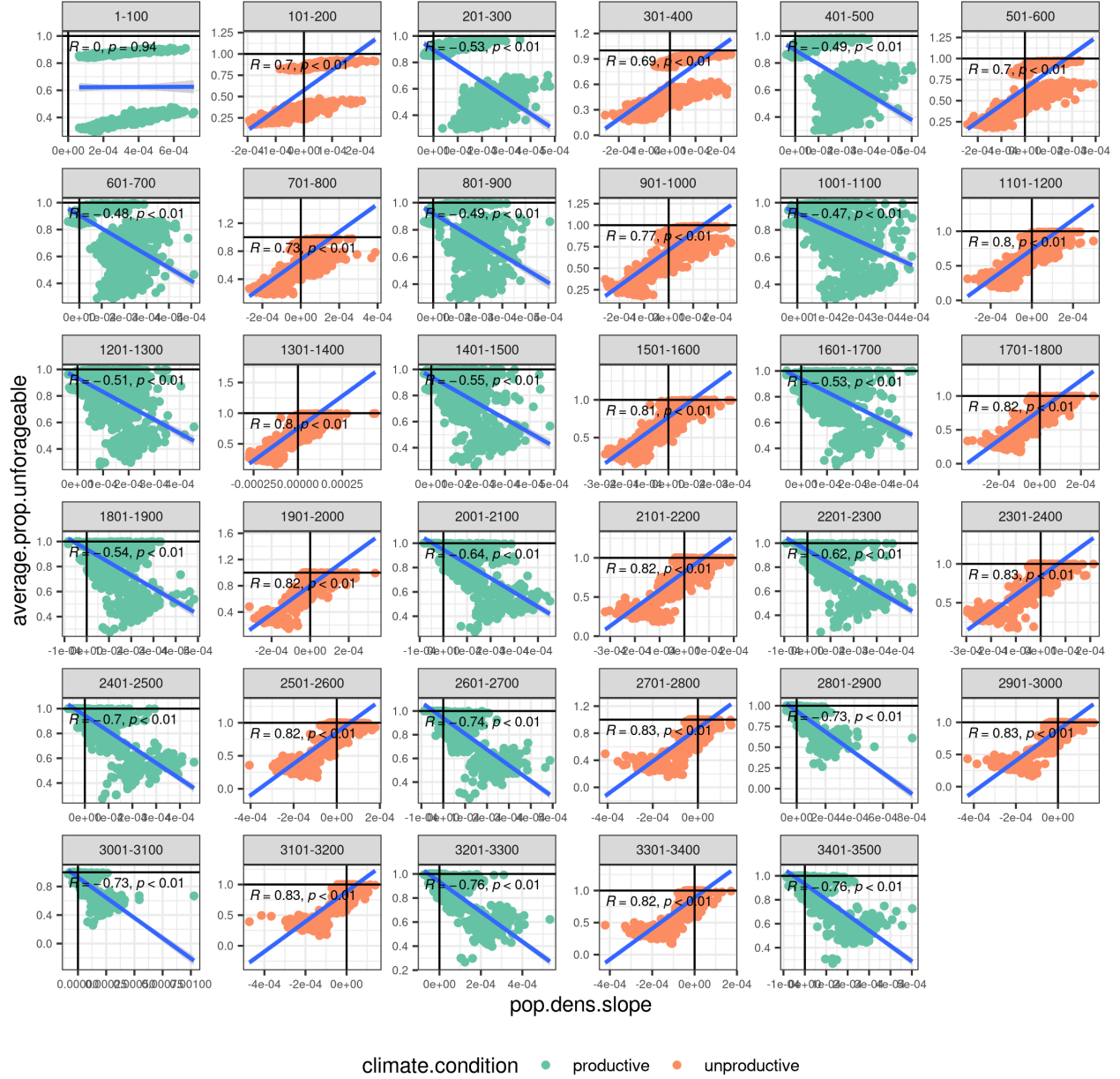


Figure S3: Relationship between change in population density within climatic interval and proportion of patches that are unforageable for short climatic intervals (100 time steps)

We can see that, with short environmental intervals, change in population density over a climatic interval is highly correlated with the proportion of patches that cannot be foraged, indicating that population is indeed controlled by access to energy (Figure S3).

Interestingly, we see opposite correlations for productive vs unproductive climatic windows. During productive periods, there will be population growth (positive slopes) with more forageable land (low proportions of unforageable patches). Conversely, during unproductive periods, there is more frequently population decline and this is strongly associated with lower proportions of unforageable land. This perhaps simply captures the fact that population decline allows for more land to be left unforaged, so smaller proportions of unforageable land, during a climatic interval. When more of the landscape is unforageable (higher proportions), we see a pattern of population stagnation in unproductive climatic intervals because there is not enough energy to allow for growth.

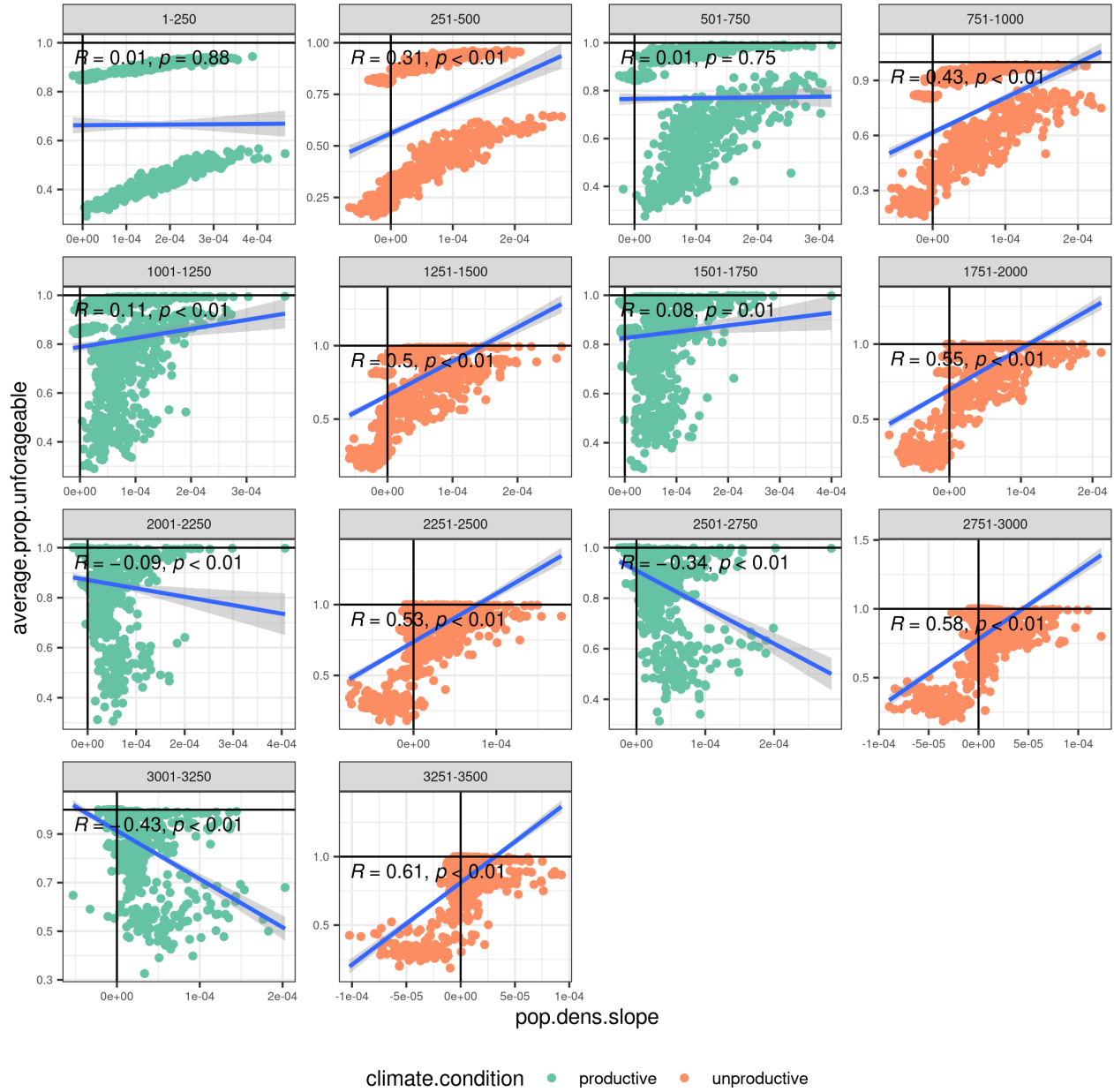


Figure S4: Relationship between change in population density within climatic interval and proportion of patches that are unforageable for long climatic intervals (250 time steps)

When climatic periods are longer (Figure S4), the significant and strong relationship between change in population density and proportions of unforageable patches is reduced or disappears in productive periods until the end of model run. This suggests that with longer climatic cycling, a different aspect of the environment is controlling population density initially.

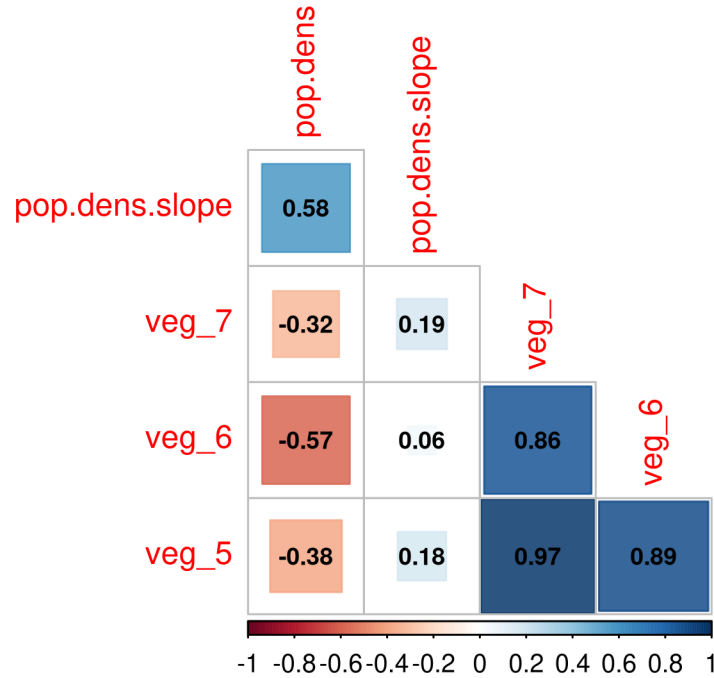


Figure S5: Correlation between population density, change in population density, and counts of high value resource types at the middle of model run during a productive climatic interval

When we look at the correlations between the slope of change in population density for a middle productive period in model runs with longer climatic cycles, we see a significant positive correlation with proportions of high vegetation types, that is patches that have a high resource value (Figure S5). This suggests that, even if most of the landscape cannot be foraged, access to a few high value patches is enough to allow for population growth. If forageable patches have lower resource values, then populations will not grow.

3 Forager probability of burning patches

We examined the effects of all model parameters on burning probability across model run using linear regressions. We model the effect of the different parameter settings on model behavior. We also model change over model run by including “ticks” as a independent variable. Finally, we investigate the interaction between time and model parameters.

```
#hist(data$mean.burn.prob)

mbp.fit1 = lm(mean.burn.prob ~ ticks + (.), data = data %>%
  dplyr::select_at(c("mean.burn.prob", "ticks", parameters)))
mbp.fit2 = lm(mean.burn.prob ~ ticks:(.), data = data %>%
  dplyr::select_at(c("mean.burn.prob", "ticks", parameters)))
mbp.fit3 = lm(mean.burn.prob ~ ticks*(.), data = data %>%
  dplyr::select_at(c("mean.burn.prob", "ticks", parameters)))

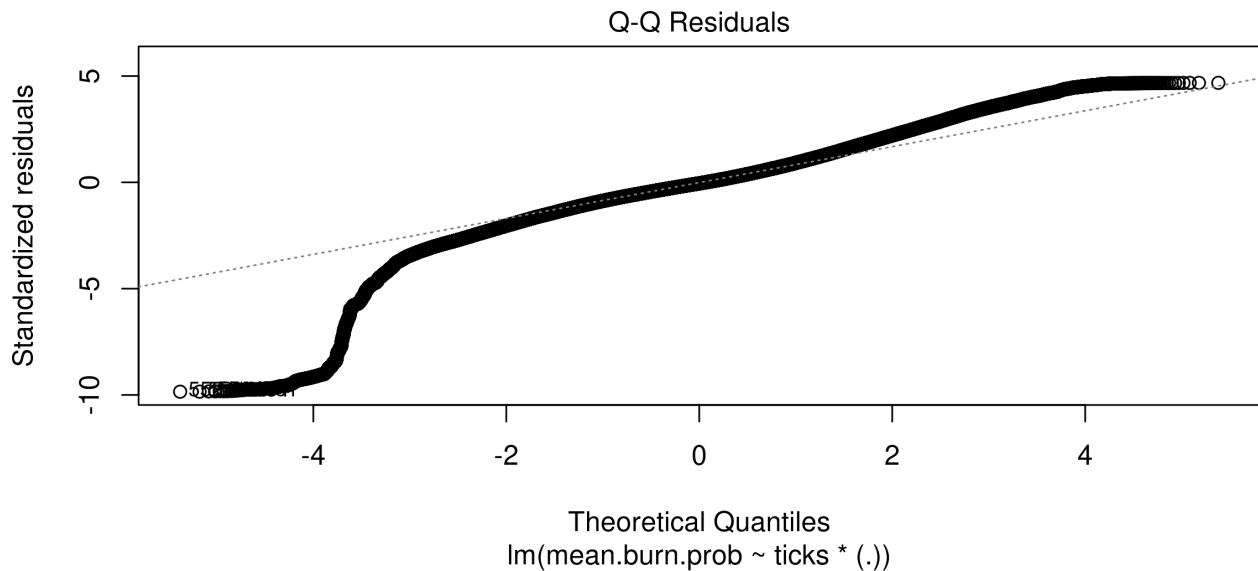
anova(mbp.fit1, mbp.fit2, mbp.fit3)

## Analysis of Variance Table
##
## Model 1: mean.burn.prob ~ ticks + (ticks + `cycle-duration` + `veg-cycle-start` +
##   `veg-distribution` + `burnt-neighbor-limit` + `burn-cost` +
##   `burn-veg-type-threshold` + `movement-model`)
```

```
## Model 2: mean.burn.prob ~ ticks:(ticks + `cycle-duration` + `veg-cycle-start` +
##   `veg-distribution` + `burnt-neighbor-limit` + `burn-cost` +
##   `burn-veg-type-threshold` + `movement-model`)
## Model 3: mean.burn.prob ~ ticks * (ticks + `cycle-duration` + `veg-cycle-start` +
##   `veg-distribution` + `burnt-neighbor-limit` + `burn-cost` +
##   `burn-veg-type-threshold` + `movement-model`)
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1 13443828 260844
## 2 13443828 93893  0    166951
## 3 13443818 86300 10     7593 118282 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

rm(list = c("mbp.fit1", "mbp.fit2"))

plot(mbp.fit3, which = 2)
```



The QQ plot shows relatively good model fit.

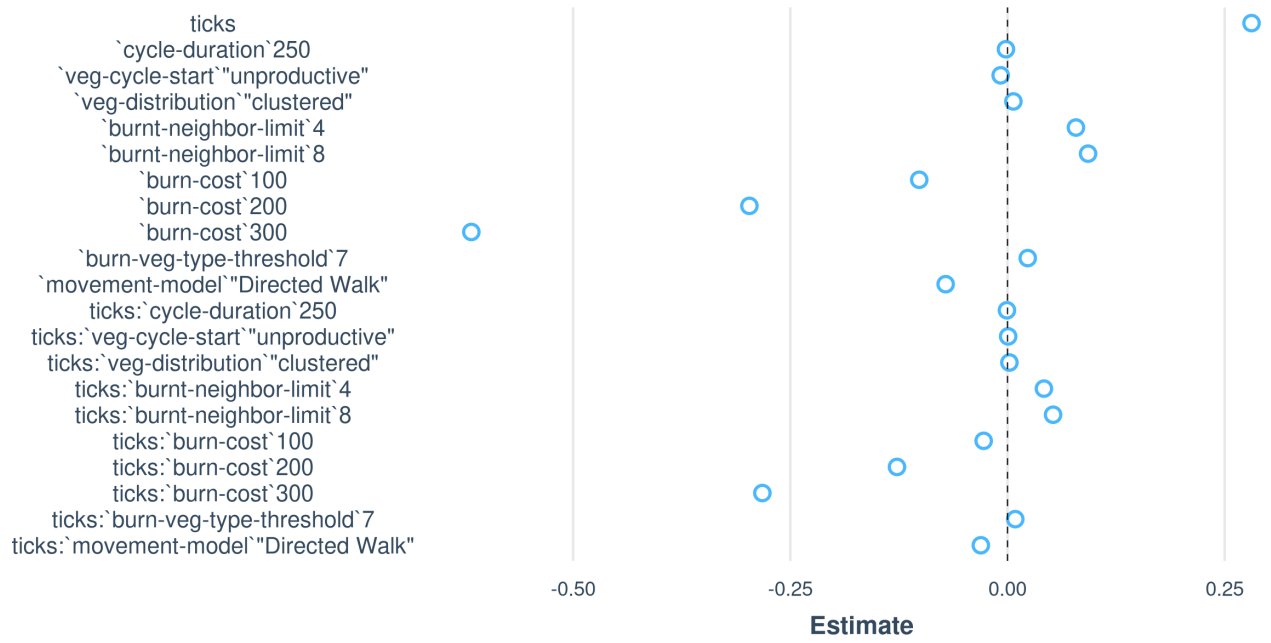


Figure S6: Standardized regression coefficients for effect of model parameters and their interactions with time (ticks) on burning probabilities

Every parameter significantly affects burning probability on average across model run (independent variables) as well as how burning probability changes during model run (interaction variables). The scaled regression variables (Figure S6) shown above also demonstrate that *burn-cost* has the strongest effect on the evolution of burning probability across model run.

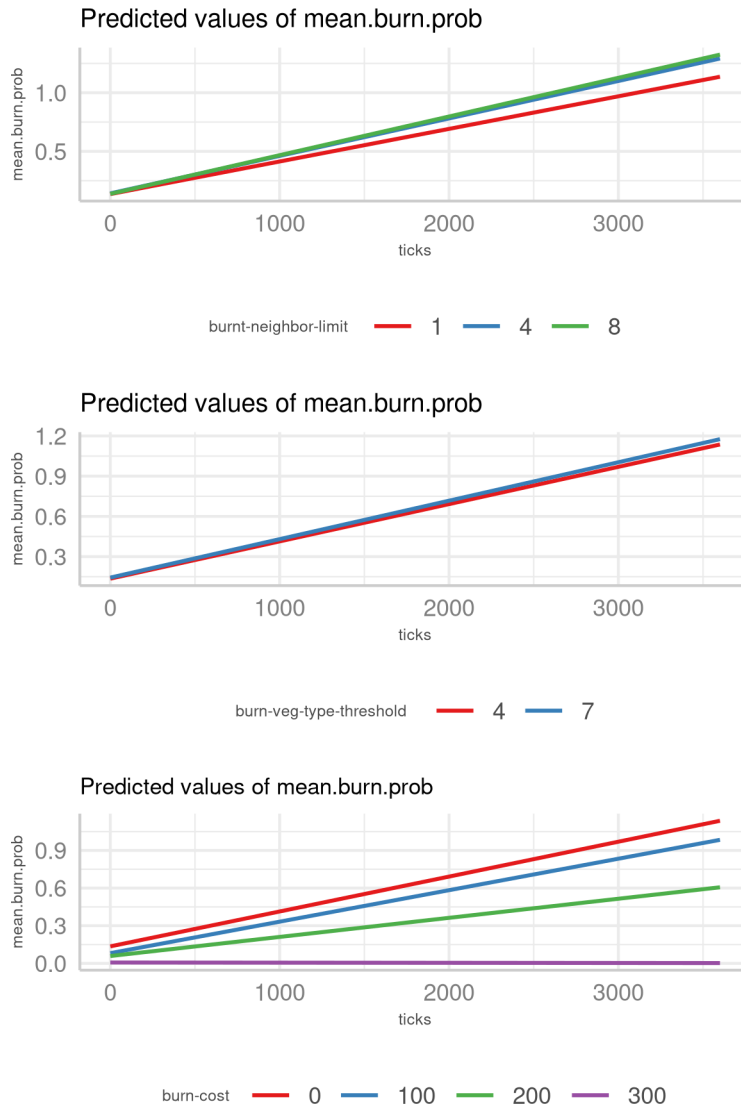


Figure S7: Predicted values of average burning probability across model run by burn cost and burn restriction parameters

In general, parameters that add limits to burning behaviors (*burn-cost*, *burnt-neighbor-limit*, *burn-veg-type-threshold*) reduce the rate of increase of burning probabilities through model run (Figure S7). The increase of burning probabilities during model run is also faster when agents move randomly.

This model behavior is confirmed by looking at the effects of model parameters on slope of the linear relationship between ticks and burning probability.

3.1 Rate of change of burning probability across model run

```
bp.rates = data.frame(
  exp = character(0),
  slope = numeric(0)
)
experiments = unique(data$exp)
for(x in experiments) {
```

```

lmdata = data %>% filter(exp == x)
fit = lm(mean.burn.prob ~ ticks, data = lmdata)
bp.rates[nrow(bp.rates) + 1, ] <- c(x, summary(fit)$coefficients[2,1])
}
bp.rates = bp.rates %>% left_join(data %>% select_at(c("exp", parameters)),
                                by = "exp", multiple = "first")
bp.rates$slope = as.numeric(bp.rates$slope)

bp.rates[,parameters] = lapply(bp.rates[,parameters], as.factor)
bp.rates$`veg-distribution` = factor(bp.rates$`veg-distribution`,
                                   levels = c("\random",
                                              "\clustered"))
bp.rates$`movement-model` = factor(bp.rates$`movement-model`,
                                   levels = c("\Random Walk",
                                              "\Directed Walk"))

length(unique((bp.rates %>% filter(slope < 0))$exp))/length(unique(bp.rates$exp))

```

```
## [1] 0.121875
```

Approximately 12% of all model runs result in a decreasing rate of change in burning probabilities.

```

neg.bprate = bp.rates %>% filter(slope < 0) %>% select_at(c(parameters)) %>%
  distinct()
table(neg.bprate$`burn-cost`)

```

```

##
##    0 100 200 300
##    0   0   0  61

```

These occur when burning costs are 300, greater than the benefit gained from burning.

```

rate.fit1 = lm(slope ~ ., data = bp.rates %>%
               select_at(c("slope", parameters)))
rate.fit2 = lm(slope ~ .:., data = bp.rates %>%
               select_at(c("slope", parameters)))
anova(rate.fit1, rate.fit2)

```

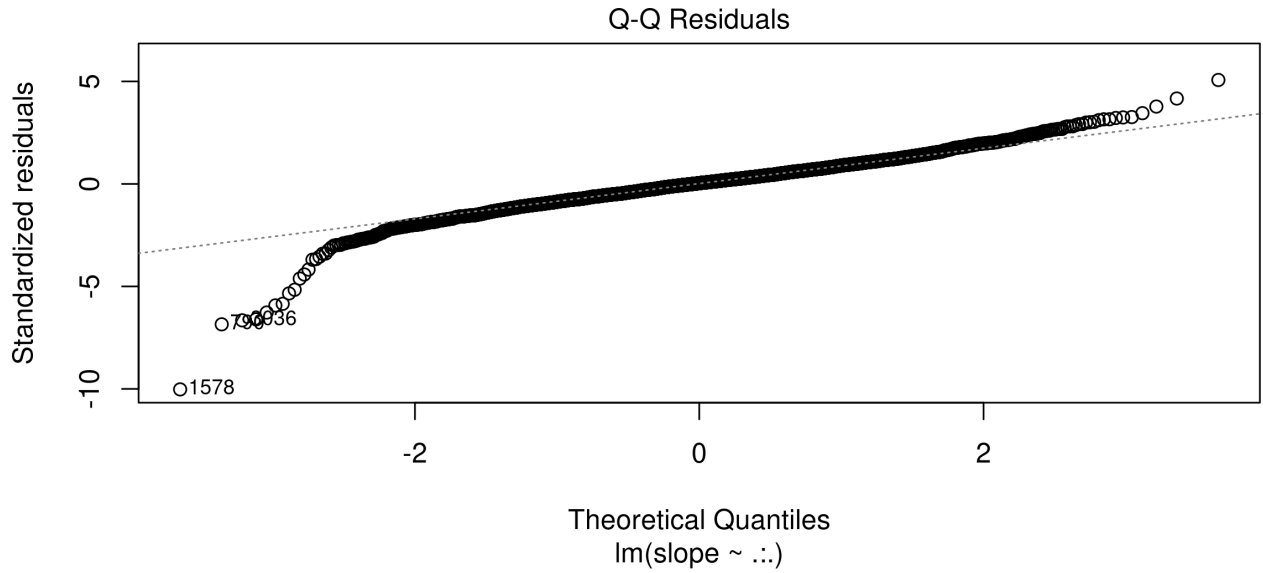
```
## Analysis of Variance Table
```

```

##
## Model 1: slope ~ `cycle-duration` + `veg-cycle-start` + `veg-distribution` +
##           `burnt-neighbor-limit` + `burn-cost` + `burn-veg-type-threshold` +
##           `movement-model`
## Model 2: slope ~ (`cycle-duration` + `veg-cycle-start` + `veg-distribution` +
##                   `burnt-neighbor-limit` + `burn-cost` + `burn-veg-type-threshold` +
##                   `movement-model`):(`cycle-duration` + `veg-cycle-start` +
##                   `veg-distribution` + `burnt-neighbor-limit` + `burn-cost` +
##                   `burn-veg-type-threshold` + `movement-model`)
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1    3829 3.6327e-06
## 2    3788 1.6139e-06 41  2.0188e-06 115.57 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
plot(rate.fit2, which = 2)
```



The QQ plot shows good model fit.



Figure S8: Standardized regression coefficients for effect of model parameters and their interactions on rate of change of burning probability across model run

The parameters that have the strongest positive impacts on the rate of change of burning behaviors are those that control the limits and costs of burning (Figure S8). Increased burning cost always decreases the rate of change, even when interacting with other variables. Increased burning costs have more negative effects when burning behaviors are less restricted by where foragers can perform burning behaviors, likely because more

burning opportunities leads to greater accrued costs. However, when burning behaviors are less restricted without a burning cost, this allows burning probabilities to increase more quickly.

3.2 Effect of self benefit on burning behaviors

Positive selection for burning behaviors across most of the model runs occurs because there is a consistent low benefit to self of these behaviors. Here, we measure benefit to self as the number of times an agent forages from a patch where they were the last agent to burn it as a proportion of the average total number of foraging events at each tick.

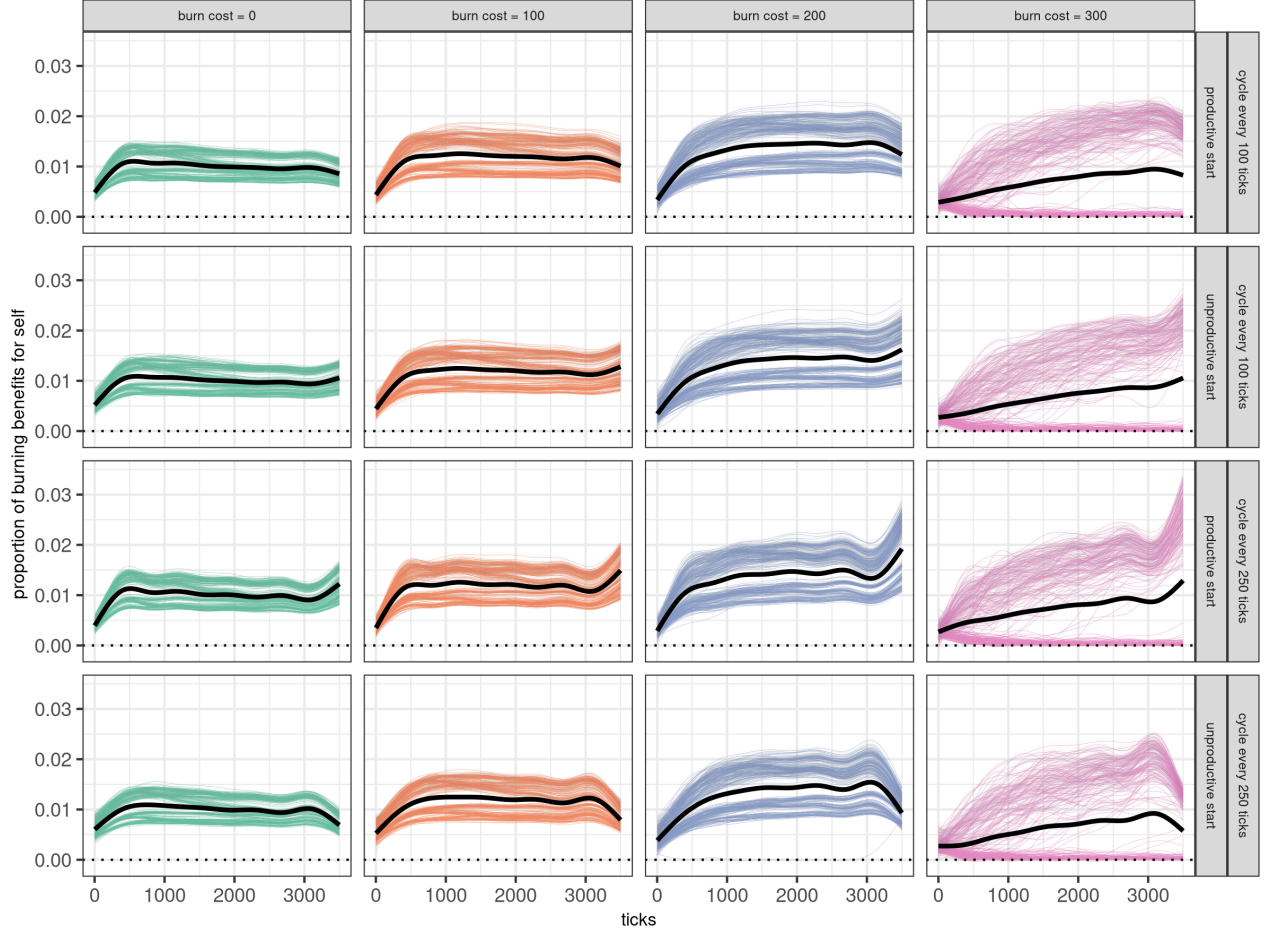


Figure S9: Proportion of foraging events where benefit of burning is recouped by agent that last performed the burning behavior at a given patch

Benefit to self consistently occurs at low levels across model run for most parameter settings, resulting in the evolution of increased burning behaviors (Figure S9).

3.2.1 Effect of model parameters on self benefit

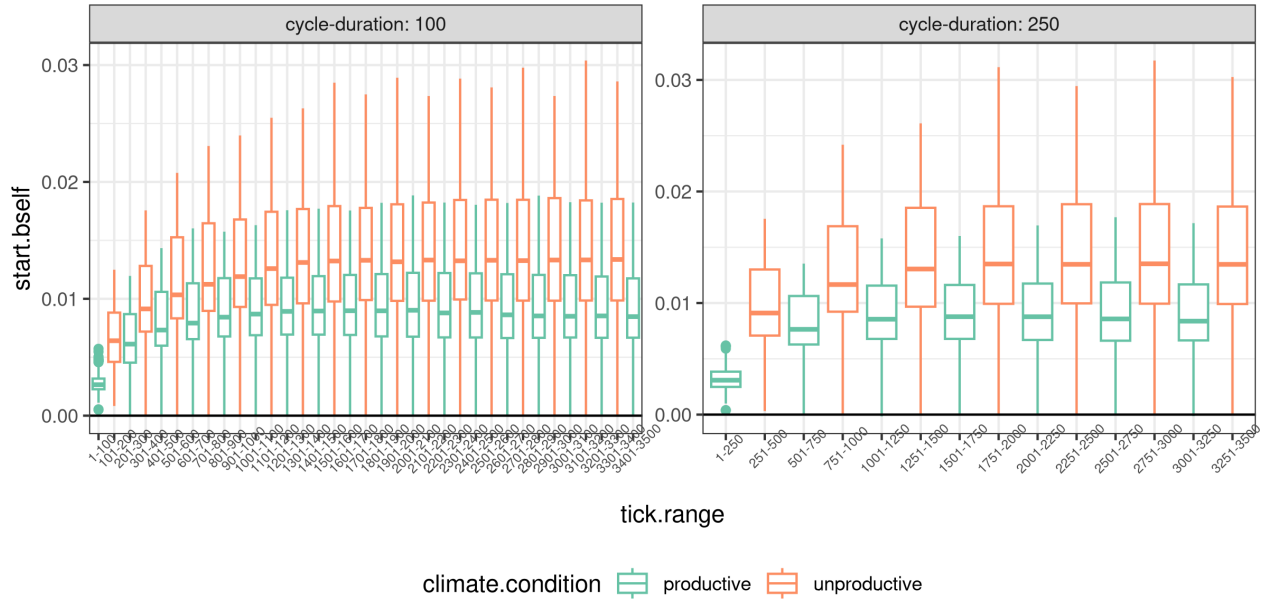


Figure S10: Average proportion of foraging events where benefit of burning is recouped by agent that last performed the burning behavior at a given patch at the start of each climatic interval during model run

When we break this down by each climatic interval, we see higher proportions of self benefit during unproductive periods (Figure S10), in part due to reduced population density in these climatic intervals.

```
fit1 = lmer(end.bself ~ start.steps
+ average.prop.unforageable
+ start.pop.density
+ (1 | tick.range),
data = rates %>%
  filter(`veg-cycle-start` == "\"productive\"") %>%
  filter(`cycle-duration` == "100") %>%
  filter(climate.condition == "productive"))
fit2 = lmer(end.bself ~ start.steps
+ average.prop.unforageable
+ start.pop.density
+ start.steps
+ (1 | tick.range),
data = rates %>%
  filter(`veg-cycle-start` == "\"productive\"") %>%
  filter(`cycle-duration` == "100") %>%
  filter(climate.condition == "unproductive"))

plot_summs(fit1, fit2, scale = T,
  model.names = c("productive intervals", "unproductive intervals"))
```

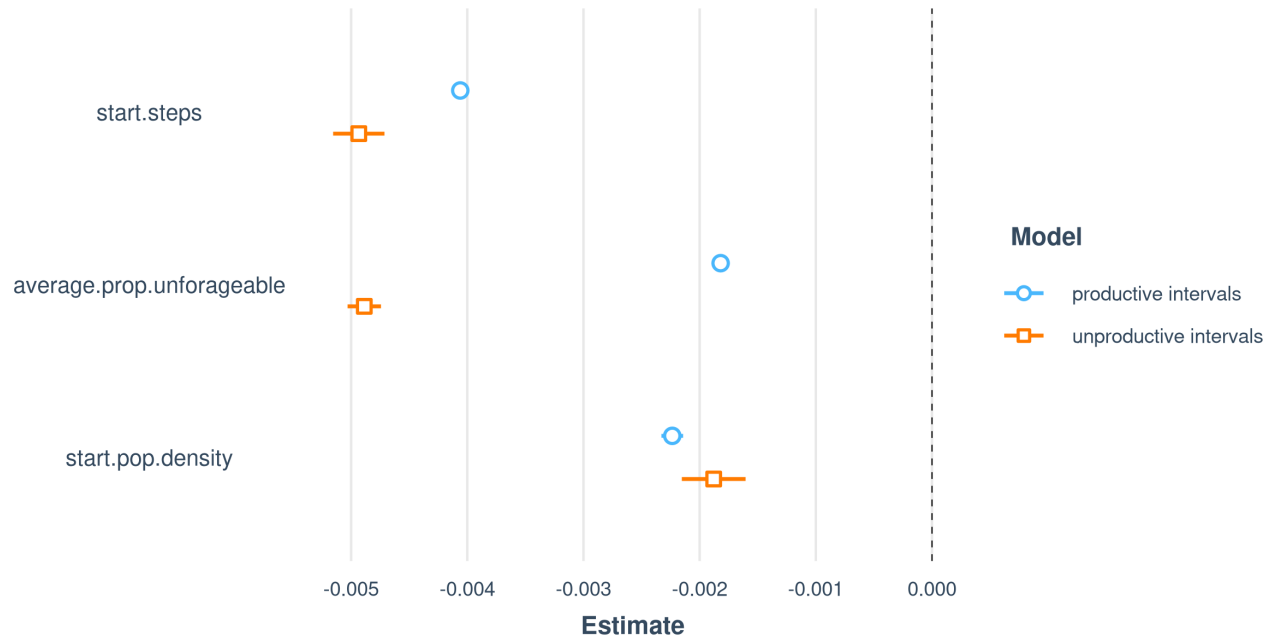


Figure S11: Standardized regression coefficients for linear mixed models on benefit to self controlling for variation within climatic intervals

This self benefit is significantly affected by forager mobility, population density, and proportion of unforageable land, as these will impact the number of patches agents have access to for foraging (Figure S11).

3.2.2 Relationship between self benefit and burning behaviors over time

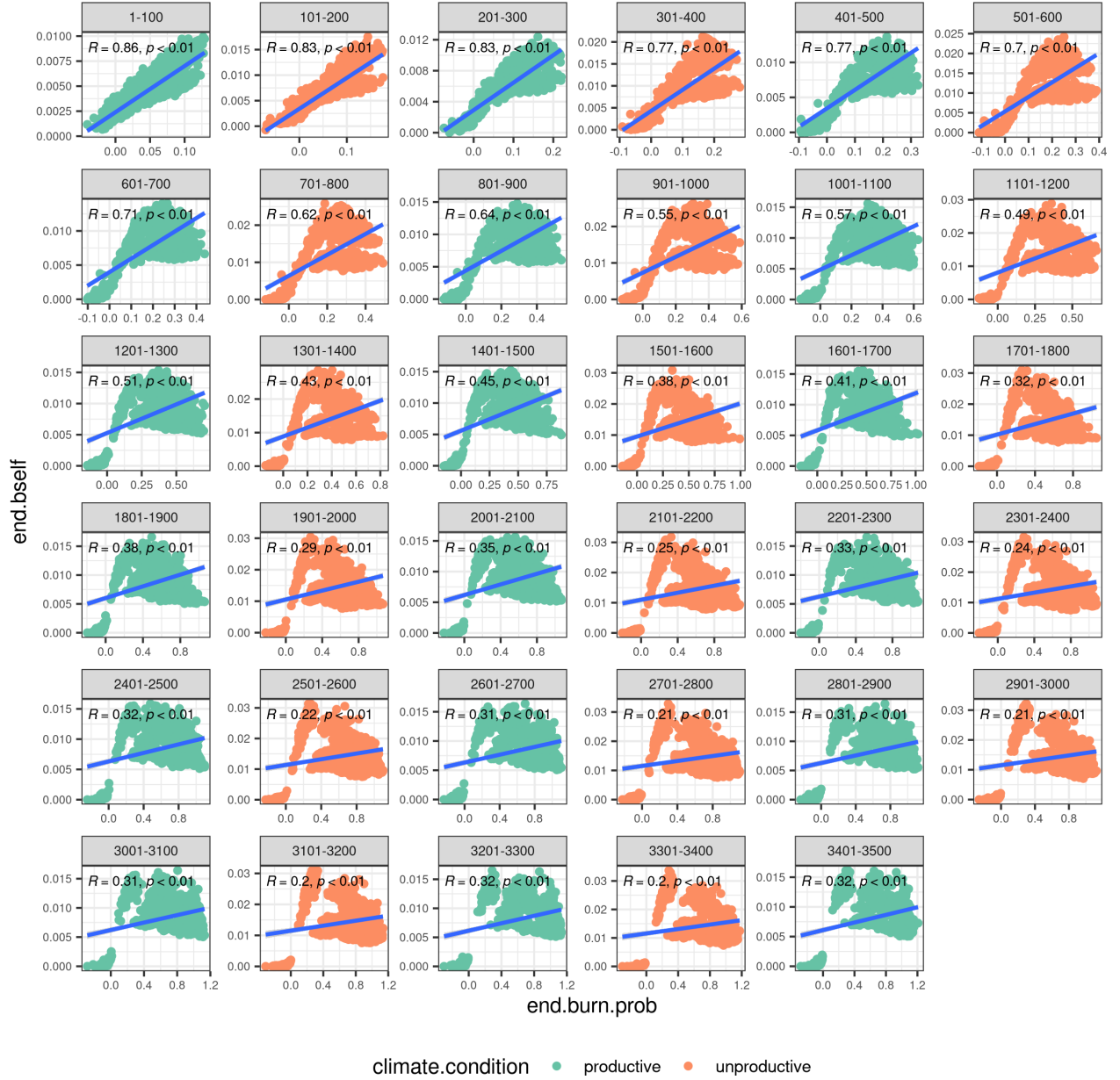


Figure S12: Relationship between proportion of self benefit and burning probability at the end of a climatic interval. Results shown for model runs that start in productive periods and have short climatic intervals (100 time steps).

When looking at how self benefit relates to burning probabilities within climatic intervals, these outputs are fairly strongly correlated at the beginning of model run (Figure S12). We can also see that, within each climatic interval, low burning probabilities cluster very tightly with low proportions of self benefit. This supports the claim that benefit to self drives positive selection for this behavior.

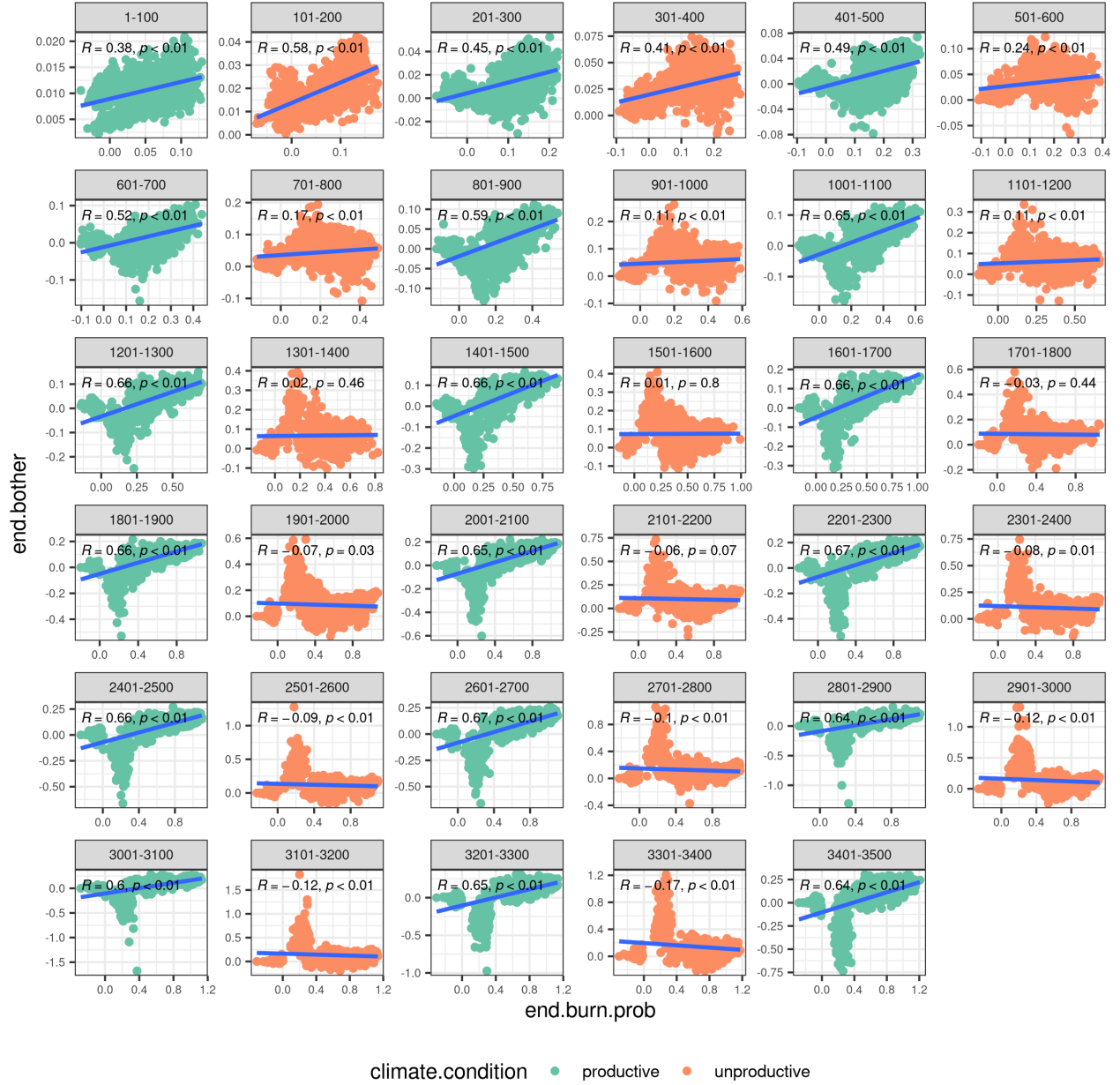


Figure S13: Relationship between proportion of other benefit (inverse of self benefit) and burning probability at the end of a climatic interval. Results shown for model runs that start in productive periods and have short climatic intervals (100 time steps).

By the end of model run, burning probabilities are more highly positively correlated with benefit to others (Figure S13), meaning that the highest probabilities of burning behaviors occur when agents besides the one that burned a patch are benefiting from the increased resource productivity. This suggests that there is a collective benefit of burning allows for further positive selection of this behavior.

3.3 Positive evolution of burning behaviors with high burning costs

One interesting model result is the evolution of high probabilities of burning under conditions where this behavior has a high cost - that is a cost that outweighs the energetic benefits gained.

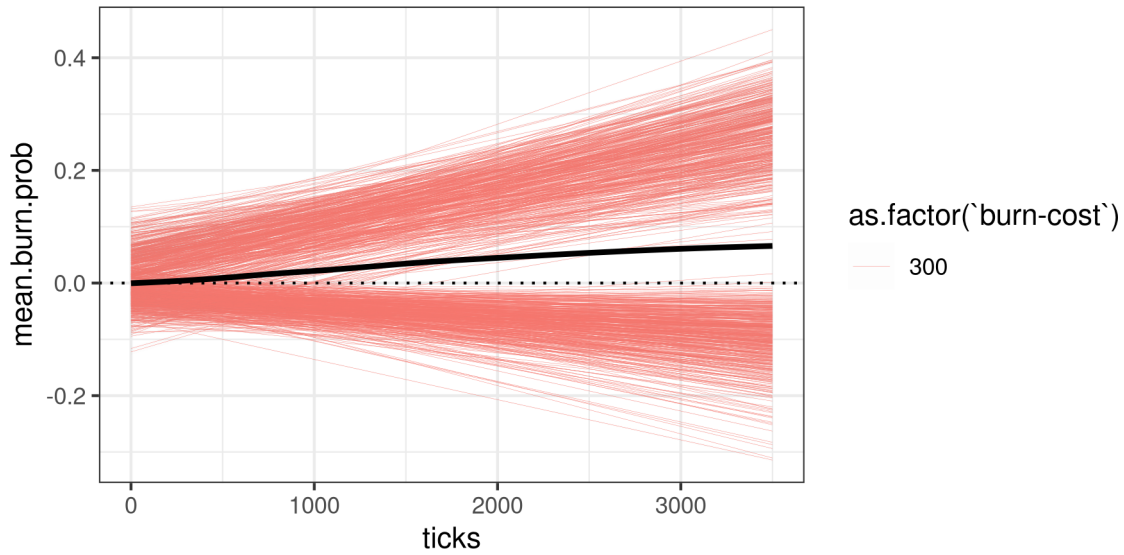


Figure S14: Trends of burning probabilities over time for model runs with high burning costs

```
bprate.300 = bp.rates %>% filter(`burn-cost` == 300)
nrow(bprate.300 %>% filter(slope < 0)) / nrow(bprate.300)
```

```
## [1] 0.4875
```

Around 49% of these high cost scenarios result in decreasing probabilities of burning behaviors (Figure S14).

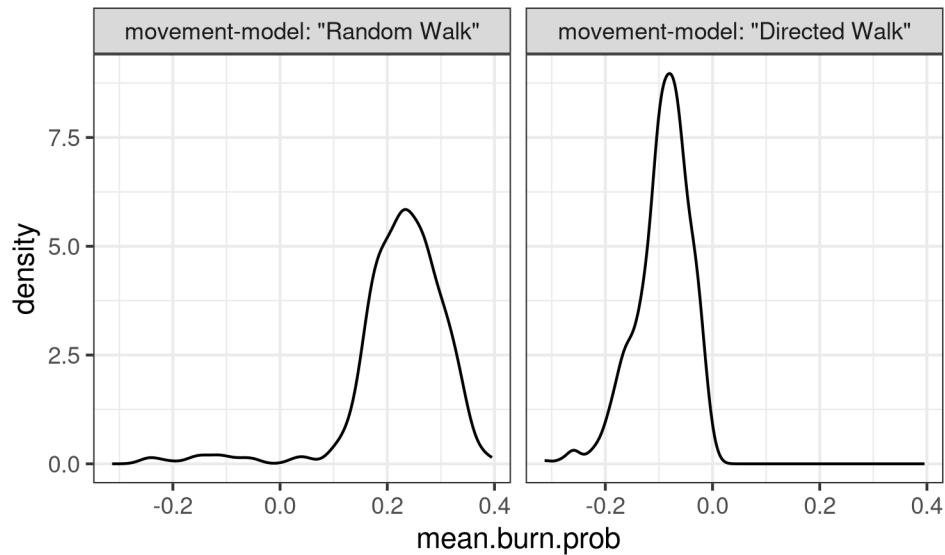


Figure S15: Burning probability at the end of the model run during high burn cost model scenarios

The final burning probability of the model during these high burn cost scenarios appears to be primarily related to the movement model that agents employ (Figure S15). Under random walks, agents are more likely to develop increased burning probabilities. Under directed walks, the high cost of burning keeps burning probabilities low or even negative. This is likely because the targeting of high yield patches allows agents to access enough energy without needing to engage in high cost behaviors. When agents have to rely on randomly encountering high value patches, it is more useful to put energy into creating an environment with more high value patches through burning behaviors.

4 Burnt landscape proportions

The proportion of burnt landscape is significantly highly correlated with population density and the average burning probability of agents (Figure S16).

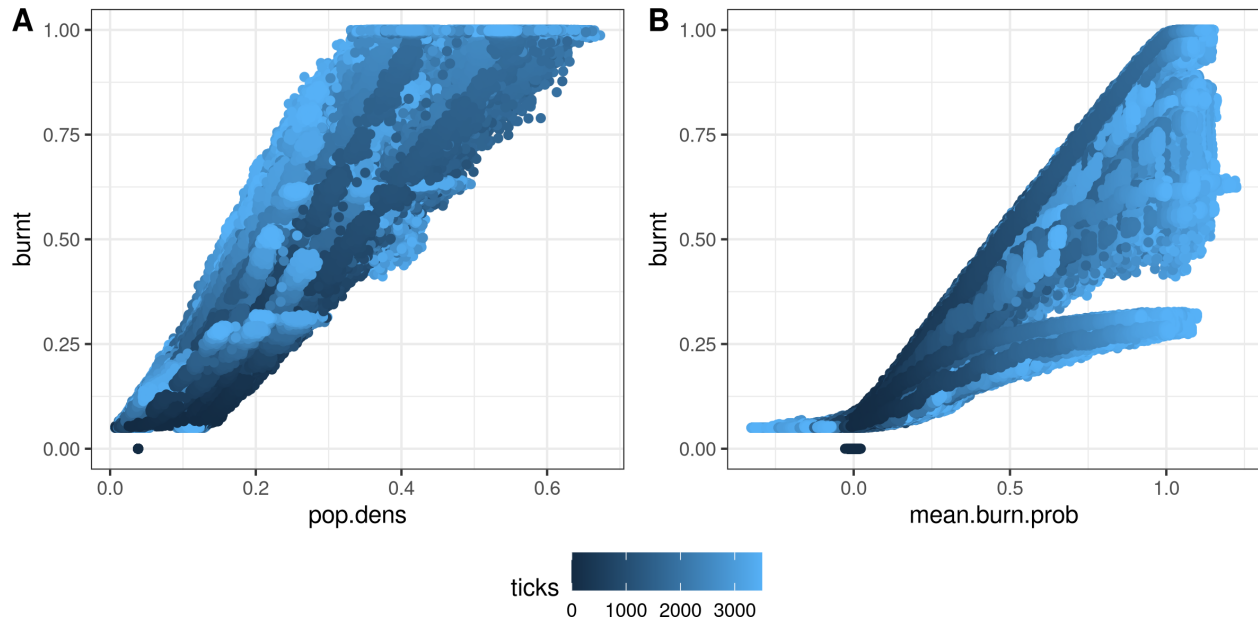


Figure S16: Relationship between proportion of landscape that is burnt and population density (A) and burning probabilities (B)

We used regressions on the proportions of burnt landscape to look at the effects of each parameter. We compared Beta regression and logistic regressions, which are often used for modeling proportion data, and linear regressions using AIC.

```
beta.data = data %>%
  mutate(burnt = ifelse(burnt == 0, burnt + 0.00001, burnt)) %>%
  mutate(burnt = ifelse(burnt == 1, burnt - 0.00001, burnt))

beta.fit1 = betareg(burnt ~ ticks*(ticks + `cycle-duration` +
  `veg-cycle-start` +
  `veg-distribution` +
  `burn-cost` +
  `burn-veg-type-threshold` +
  `burnt-neighbor-limit` +
  `movement-model`),
  data = beta.data %>%
    select_at(c("ticks", "burnt", parameters)))

log.fit1 = glm(burnt ~ ticks*(ticks + `cycle-duration` +
  `veg-cycle-start` +
  `veg-distribution` +
  `burn-cost` +
  `burn-veg-type-threshold` +
  `burnt-neighbor-limit` +
  `movement-model`),
  data = data %>%
```

```

      select_at(c("ticks", "burnt", parameters)),
      family = binomial)

lin.fit1 = lm(burnt ~ ticks*(ticks + `cycle-duration` +
      `veg-cycle-start` +
      `veg-distribution` +
      `burn-cost` +
      `burn-veg-type-threshold` +
      `burnt-neighbor-limit` +
      `movement-model`),
      data = data %>% select_at(c("ticks", "burnt", parameters)))

AIC(beta.fit1)

## [1] -30663594
AIC(log.fit1)

## [1] 7754388
AIC(lin.fit1)

## [1] -26381640
AIC(beta.fit1) < AIC(lin.fit1)

## [1] TRUE

```

Comparison of AIC shows that the Beta regression performs best.

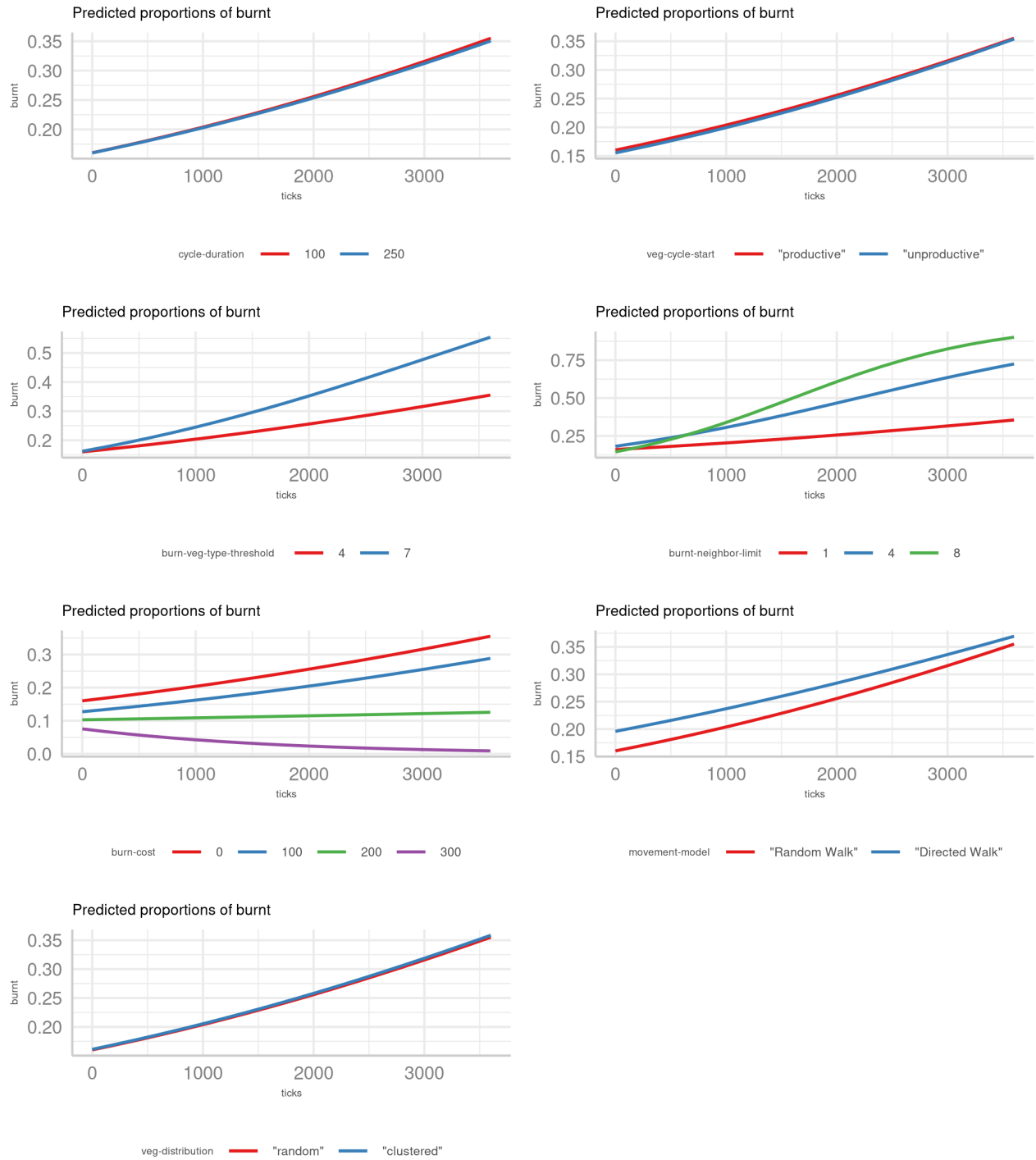


Figure S17: Predicted proportions of landscape that are burnt across model run for different model parameters

The predicted proportions of burnt landscape from the best fit Beta regression model show that increasing limits on location of burning or increasing cost to burning significantly affect the amount of burnt landscape (Figure S17). This matches the results above for the effect of these parameters on mean burning probability. This is also consistent with the high positive correlation between proportion of burnt landscape and burning probability through time.

5 Spatial clustering of vegetation types

Spatial clustering (and its converse spatial heterogeneity) of vegetation types is assessed via Moran's I values. We limit our analysis to those significant values of Moran's I to assess the degree of clustering (positive values) or dispersion (negative values). We assessed the effects of the different parameters using linear regressions.

```
# hist(veg.100.prod$morans.i)
# hist(veg.100.unprod$morans.i)
# hist(veg.250.prod$morans.i)
# hist(veg.250.unprod$morans.i)

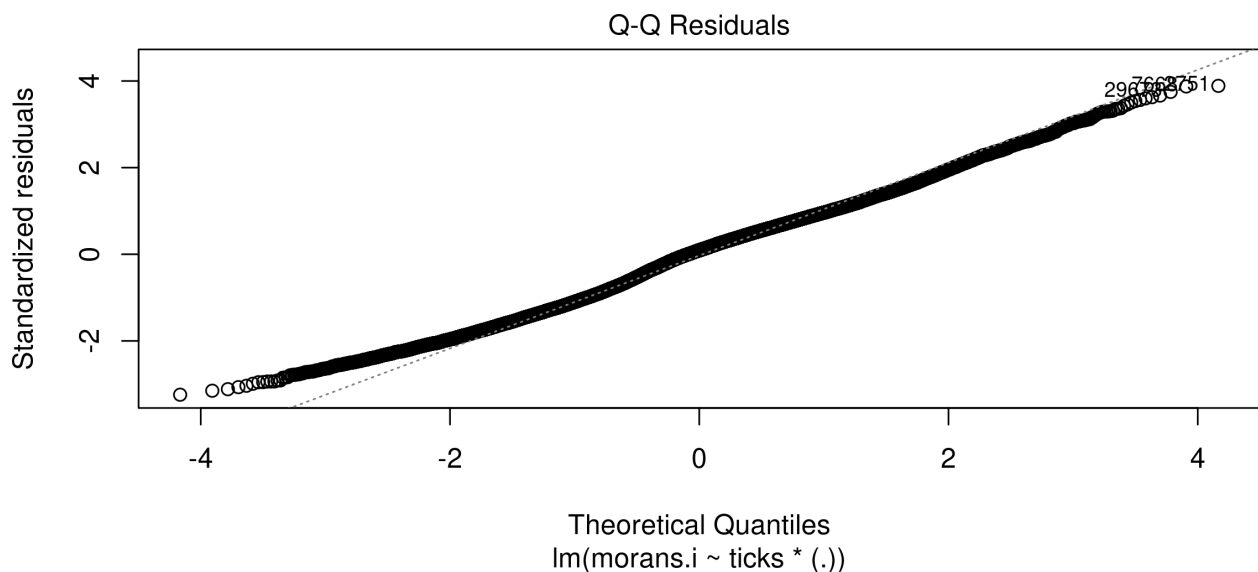
veg.100.prod.fit1 = lm(morans.i ~ ticks*(.),
                      data = veg.100.prod %>%
                        filter(signif == T) %>%
                        select_at(c("ticks", "morans.i", reg.params)))

veg.100.unprod.fit1 = lm(morans.i ~ ticks*(.),
                        data = veg.100.unprod %>%
                          filter(signif == T) %>%
                          select_at(c("ticks", "morans.i", reg.params)))

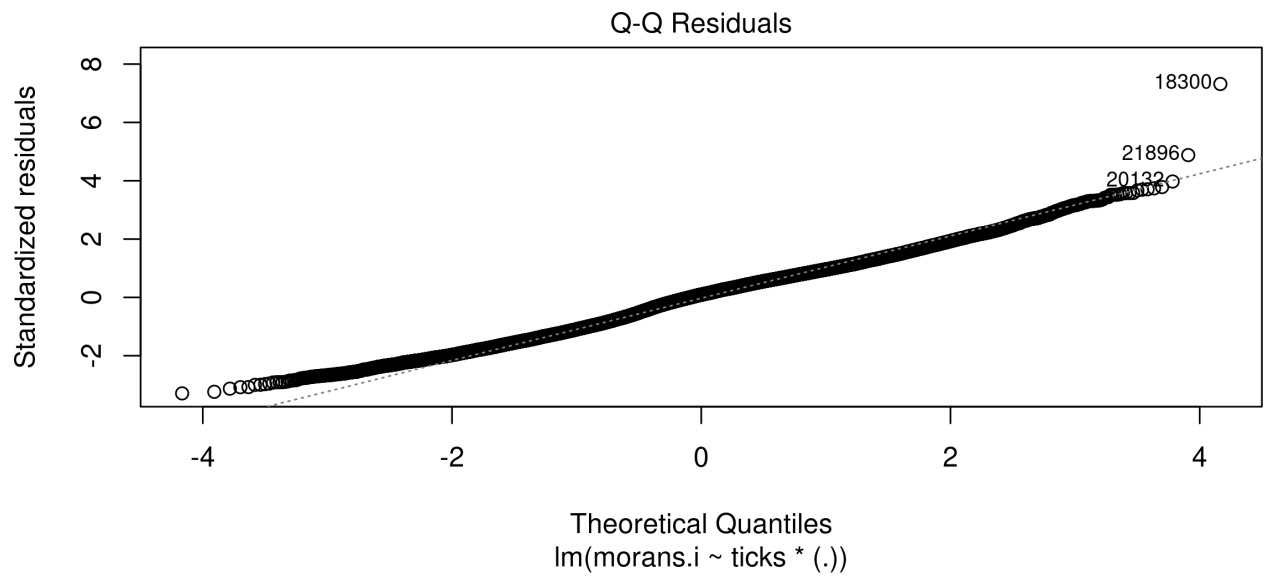
veg.250.prod.fit1 = lm(morans.i ~ ticks*(.),
                      data = veg.250.prod %>%
                        filter(signif == T) %>%
                        select_at(c("ticks", "morans.i", reg.params)))

veg.250.unprod.fit1 = lm(morans.i ~ ticks*(.),
                        data = veg.250.unprod %>%
                          filter(signif == T) %>%
                          select_at(c("ticks", "morans.i", reg.params)))

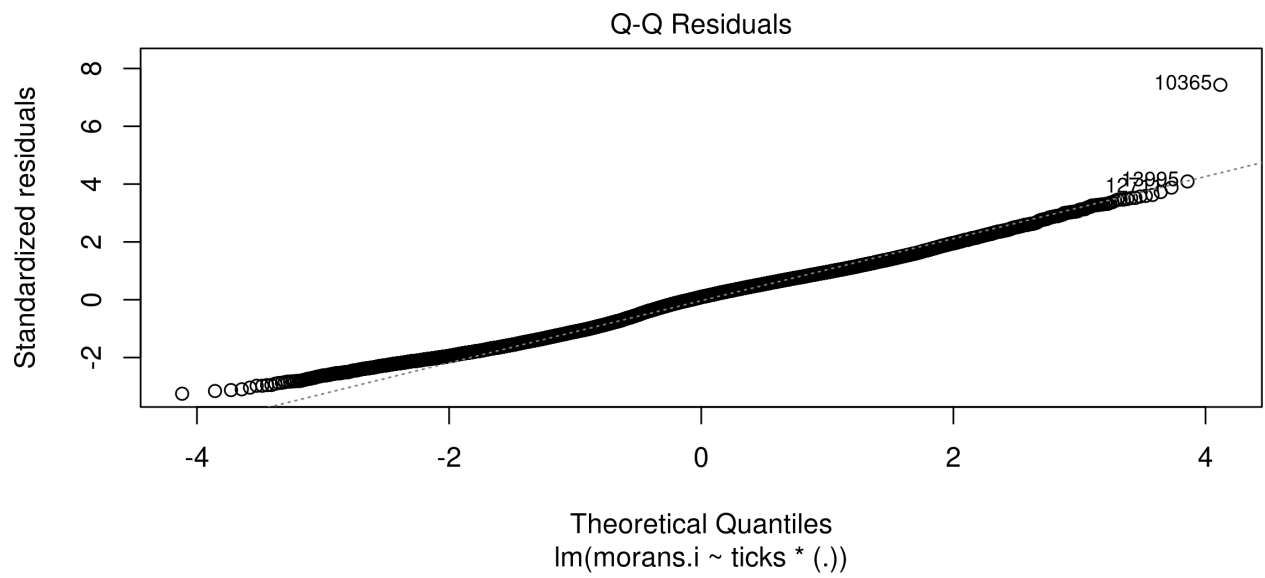
plot(veg.100.prod.fit1, which = 2)
```



```
plot(veg.100.unprod.fit1, which = 2)
```



```
plot(veg.250.prod.fit1, which = 2)
```



```
plot(veg.250.unprod.fit1, which = 2)
```

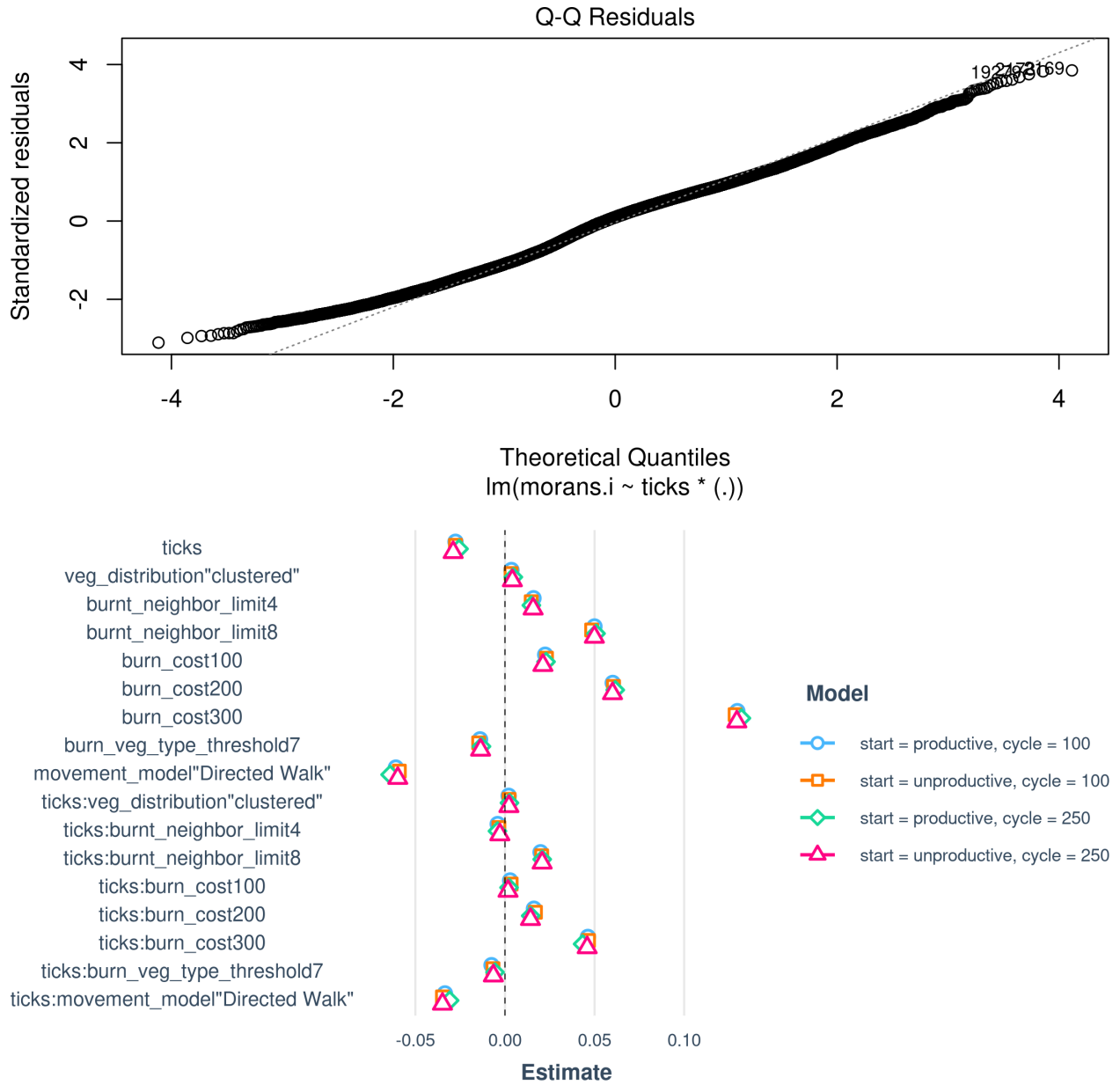


Figure S18: Standardized regression coefficients of model parameters and their interaction with time on global spatial autocorrelation of vegetation types. Results shown for four different model scenarios based on climatic interval length and starting interval type

Looking at the standardized regression coefficients of the linear regressions, we can see that high/medium cost of burning behaviors and a directed walk movement model have the greatest impact on clustering of vegetation types (Figure S18). Interestingly, the underlying distribution of maximum potential vegetation types for patches has essentially no impact on spatial clustering of vegetation types relative to the other parameters.

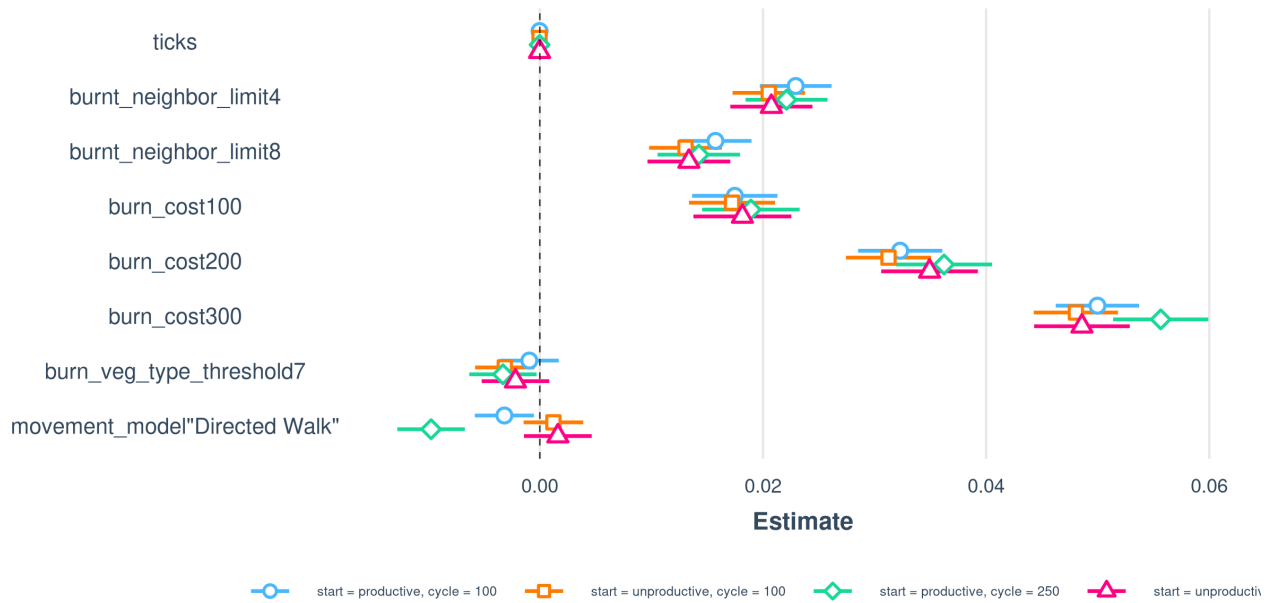


Figure S19: Unstandardized regression coefficients of model parameters and their interaction with time on global spatial autocorrelation of vegetation types. Results shown for four different model scenarios based on climatic interval length and starting interval type

We compare unstandardized parameter effects for the most impactful parameters, we can see that most of the parameters have similar effects regardless of the cycle duration or starting interval type (Figure S19). The parameter with significant differences in regression coefficients between the models is: *movement-model*. We can see that when model run starts in a productive state and the climatic cycles are longer, Directed Walks allow for, on average, less spatial clustering of vegetation types.

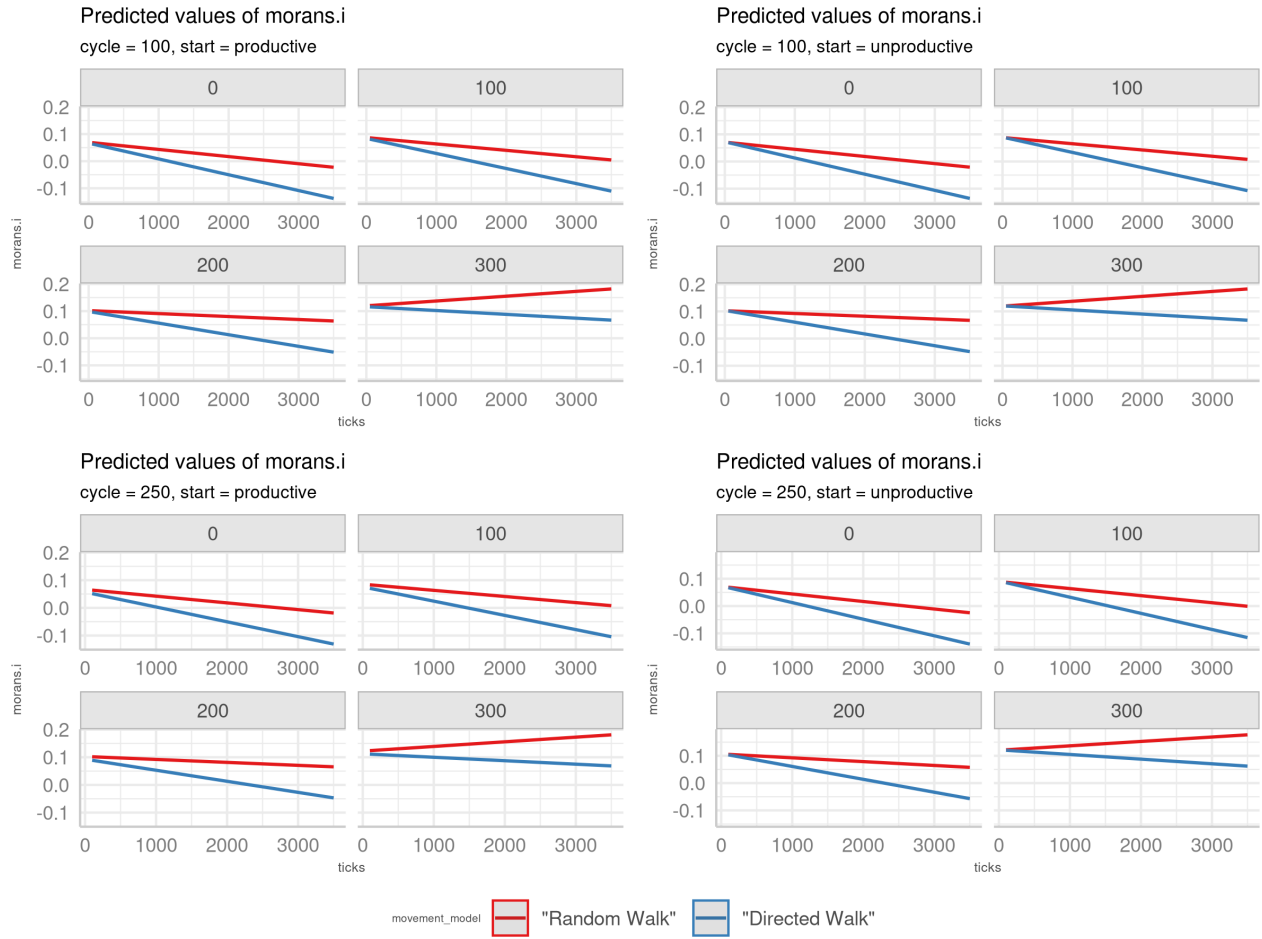


Figure S20: Predicted values of spatial autocorrelation of vegetation types over model run by burning cost (grey facets) and movement models

When we look at how vegetation clustering changes over time with different burning costs (facets) and different movement models, it is clear that directed walks result in a faster rate of decline in spatial clustering than random walks across all burning costs (Figure S20). When burning costs are low/none, this decrease is even steeper as more burning occurs, with spatial autocorrelation becoming more dispersed, at negative values, by the end of model run. This is consistent with above results that directed walk result in faster rates of increase in burning probabilities and more burnt patches over time.

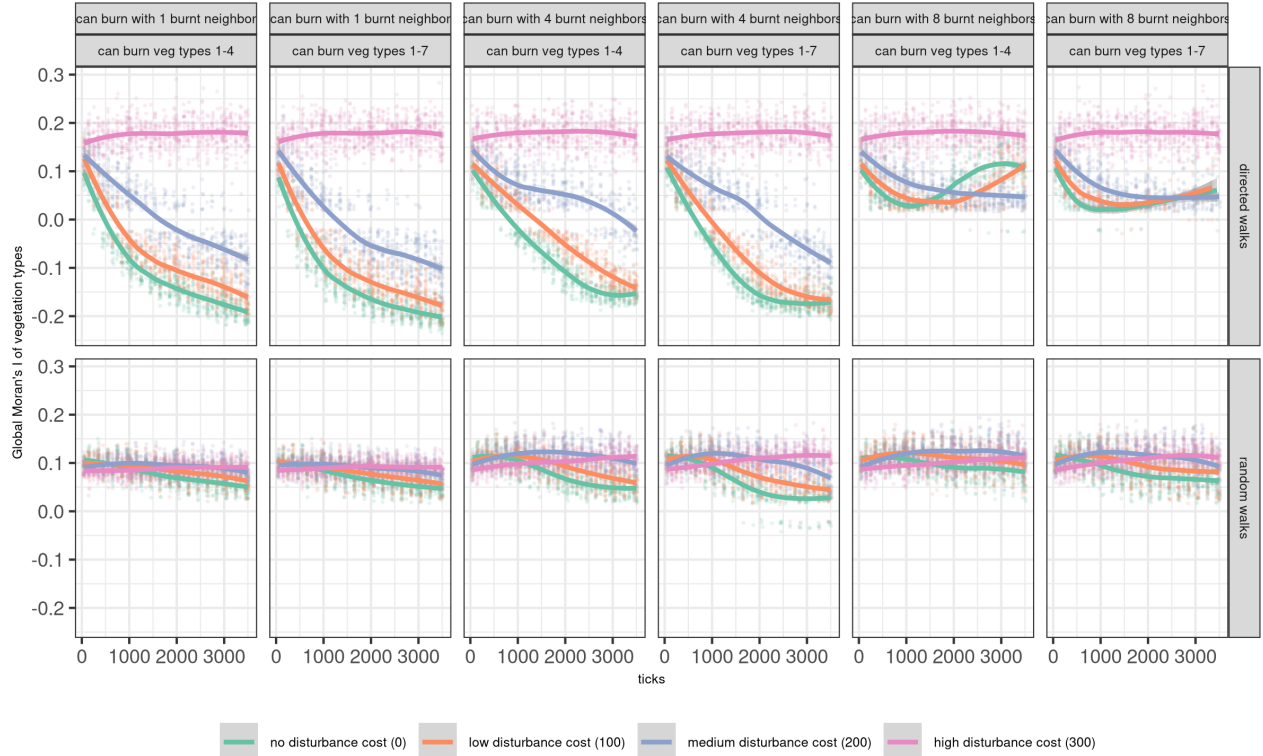


Figure S21: Trends in spatial autocorrelation of vegetation types across model run by parameters for burning costs, movement models and restrictions of burning behaviors

There are also differing trends based on spatial limits and vegetation limits on burning behaviors. When patches can only be burnt with 4 or less burnt neighbors, spatial clustering of vegetation types is much more decreased (Figure S21), particularly for lower disturbance costs that allow for more burning.

6 Vegetation diversity

Vegetation diversity was measured by Shannon's Diversity Index. This metric considers the proportional abundances of vegetation types to indicate whether one type dominates or abundances are more evenly spread across all types. A SDI value of 0 indicates a lack of diversity with abundance being concentrated in a single species. Higher SDI values indicate more diversity of vegetation types.

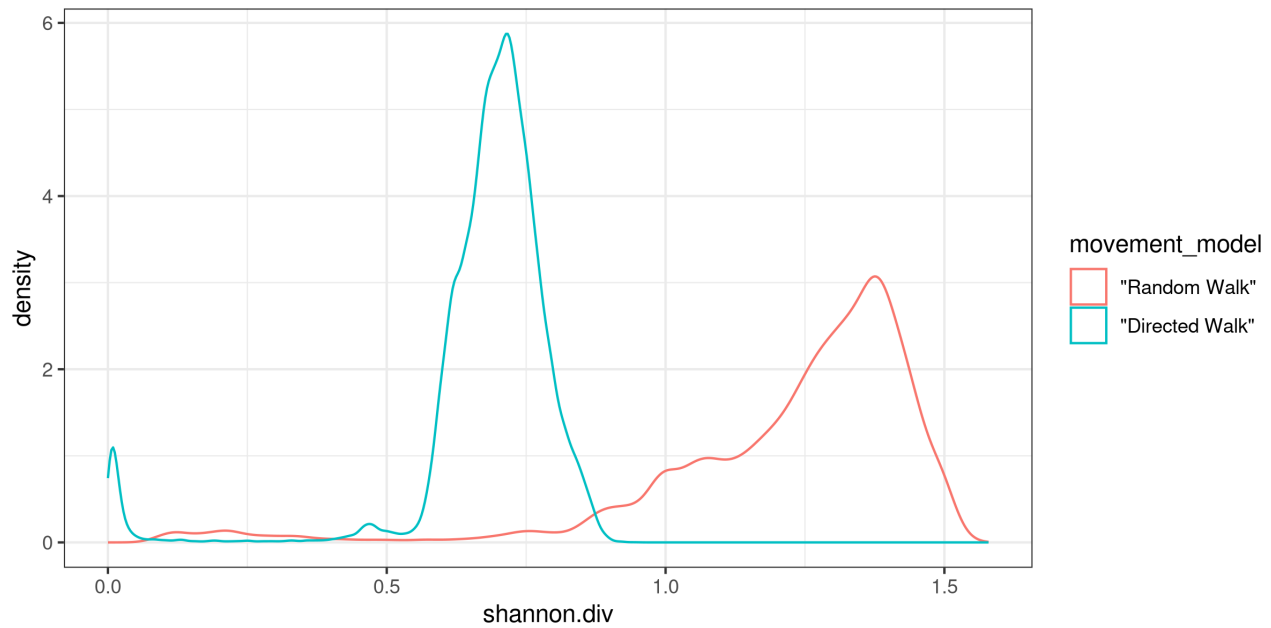


Figure S22: Distribution of vegetation diversity by movement model

The distribution of this metric is bimodal across model runs. This bimodality is caused by the different movement models used: directed walks result in less diversity than random walks (Figures S22).

6.1 Vegetation diversity under random walks

We fit a GAM to vegetation type diversity for model runs with random walks because it is a better fit than a linear model. We use GAM to investigate how the effects of the parameters change through time.

```
veg.data.r = veg.data %>% filter(movement_model == "\"Random Walk\"")
reg.params = parameters = c(
  "cycle_duration", # 100, 250
  "veg_cycle_start", # productive, unproductive
  "veg_distribution", # random, clustered
  "burnt_neighbor_limit", # 1, 4, 8
  "burn_cost", # 0, 50
  "burn_veg_type_threshold"
)
veg.data.r[,reg.params] = lapply(veg.data.r[,reg.params], as.factor)

rw.gam1 = gam(shannon.div ~
  #s(ticks, by = natural_ignition) +
  s(ticks, by = veg_cycle_start) +
  s(ticks, by = veg_distribution) +
  s(ticks, by = burn_cost) +
  s(ticks, by = burn_veg_type_threshold) +
  s(ticks, by = burnt_neighbor_limit) +
  s(ticks, by = cycle_duration), data = veg.data.r)

rw.gam2 = gam(shannon.div ~ veg_cycle_start +
  veg_distribution +
  burn_cost +
  burn_veg_type_threshold +
```

```

burnt_neighbor_limit +
cycle_duration +
#s(ticks, by = natural_ignition) +
s(ticks, by = veg_cycle_start) +
s(ticks, by = veg_distribution) +
s(ticks, by = burn_cost) +
s(ticks, by = burn_veg_type_threshold) +
s(ticks, by = burnt_neighbor_limit) +
s(ticks, by = cycle_duration), data = veg.data.r)

anova(rw.gam1, rw.gam2, test = "Chisq")

## Analysis of Deviance Table
##
## Model 1: shannon.div ~ s(ticks, by = veg_cycle_start) + s(ticks, by = veg_distribution) +
##   s(ticks, by = burn_cost) + s(ticks, by = burn_veg_type_threshold) +
##   s(ticks, by = burnt_neighbor_limit) + s(ticks, by = cycle_duration)
## Model 2: shannon.div ~ veg_cycle_start + veg_distribution + burn_cost +
##   burn_veg_type_threshold + burnt_neighbor_limit + cycle_duration +
##   s(ticks, by = veg_cycle_start) + s(ticks, by = veg_distribution) +
##   s(ticks, by = burn_cost) + s(ticks, by = burn_veg_type_threshold) +
##   s(ticks, by = burnt_neighbor_limit) + s(ticks, by = cycle_duration)
##   Resid. Df Resid. Dev      Df Deviance Pr(>Chi)
## 1      64261      2746.9
## 2      64247      1598.0 13.392   1148.9 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

rw.fit1 = lm(shannon.div ~ ticks*(.), data = veg.data.r %>%
  select_at(c("ticks", "shannon.div", reg.params)))

AIC(rw.gam2)

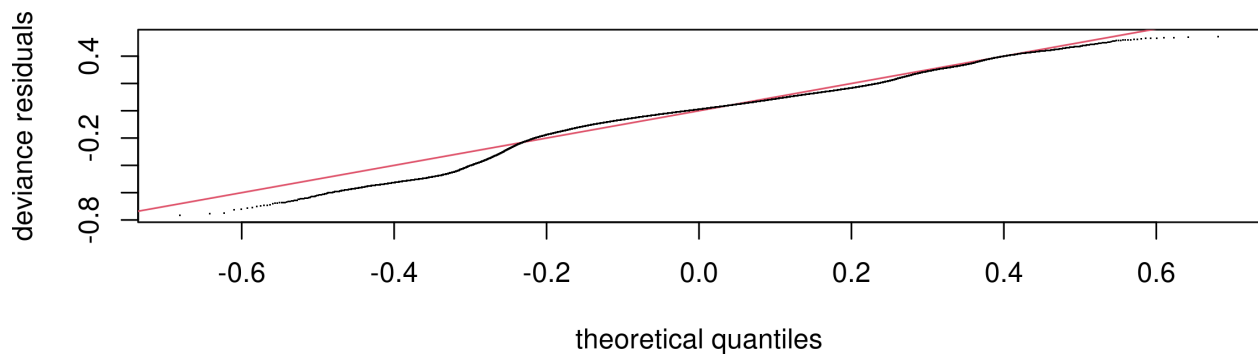
## [1] -55002.36

AIC(rw.fit1)

## [1] -48005.73

qq.gam(rw.gam2)

```



The QQ plot of this fit shows good model performance.

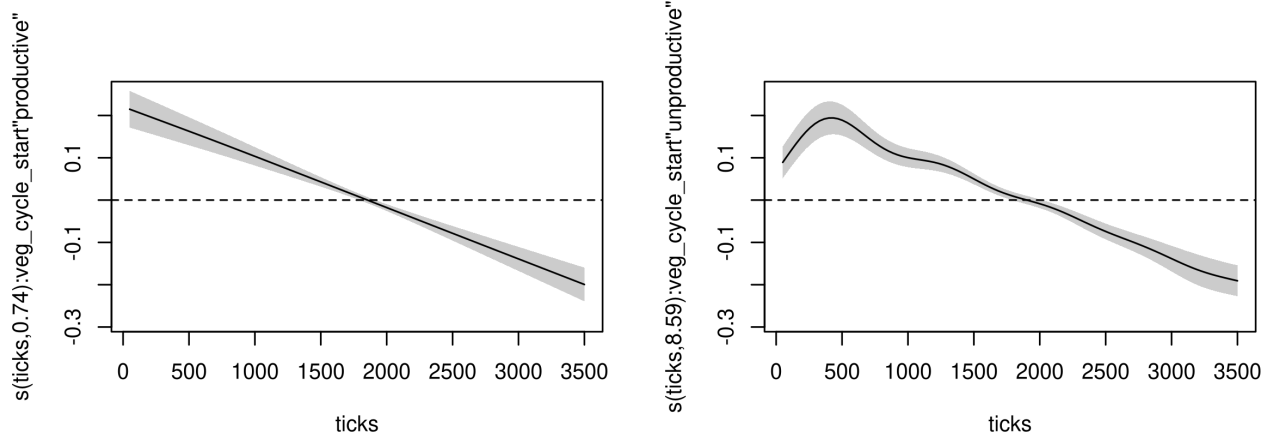


Figure S23: Effect through time of type of starting climatic period on vegetation diversity

The starting environmental conditions do not appear to have significantly differing effects on the evolution of vegetation diversity over model run (Figure S23).

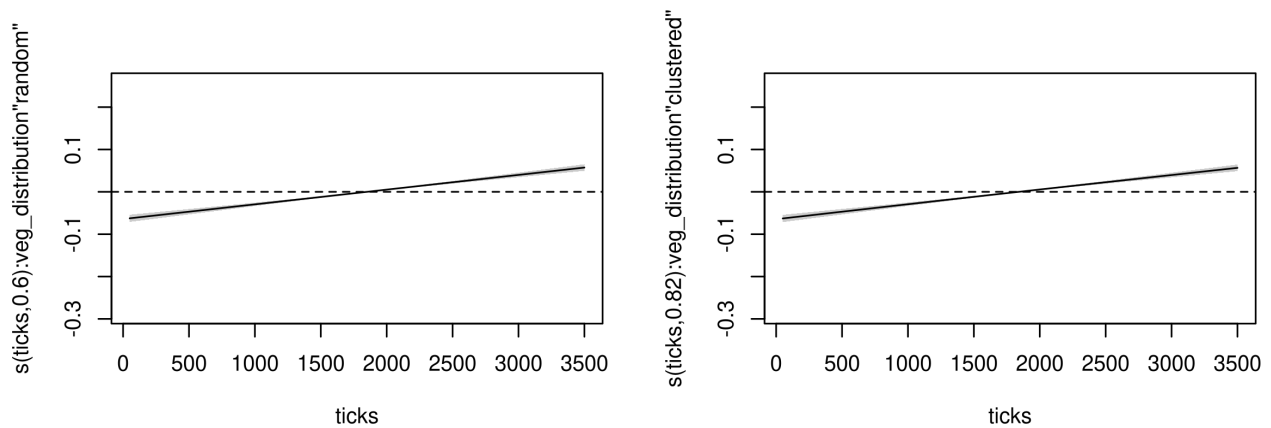


Figure S24: Effect through time of underlying distribution of vegetation types on vegetation diversity

The underlying distribution of resource potential, or maximum vegetation type value, also do not have different effects on vegetation diversity - both distributions have slightly increasingly positive effects over the course of model run (Figure S24).

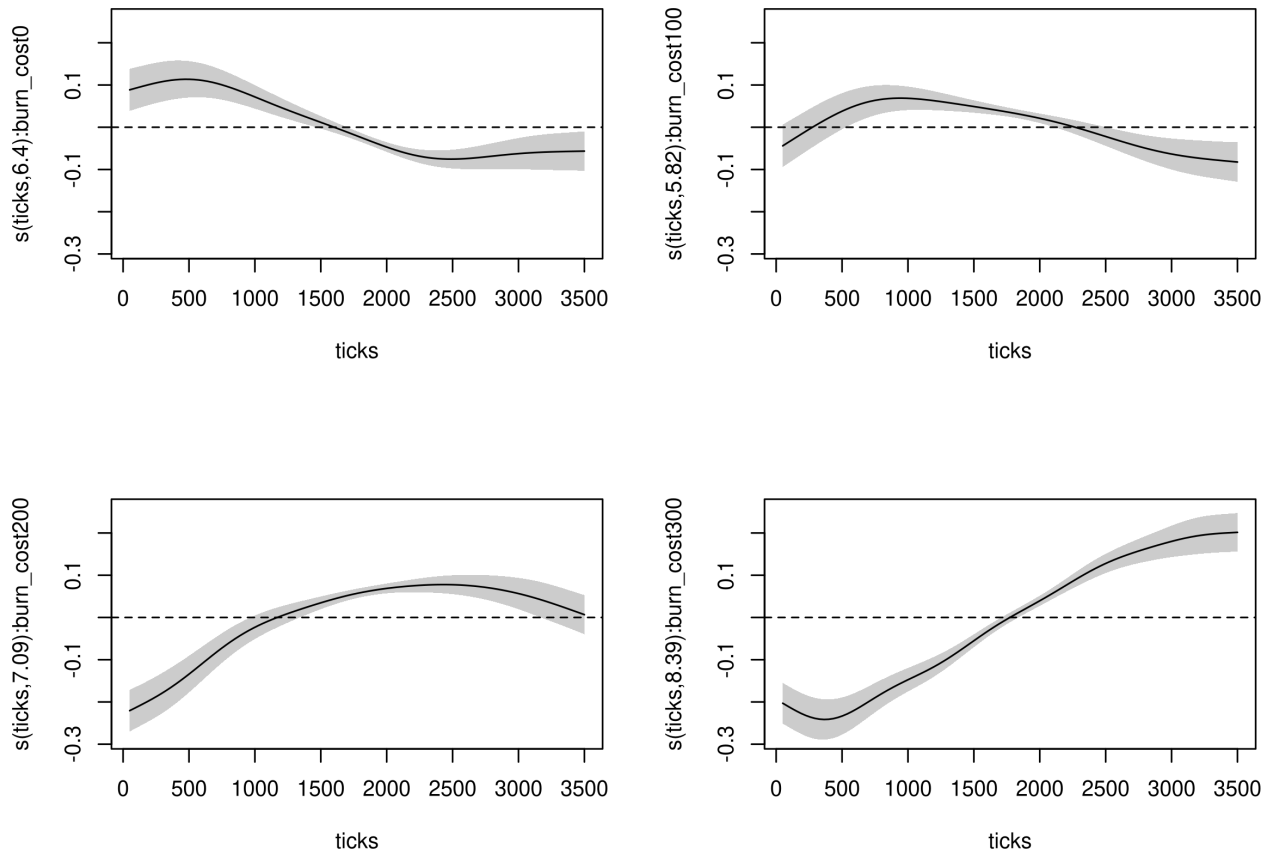


Figure S25: Effect through time of burn cost on vegetation diversity

Having a cost associated with burning introduces a larger effect on vegetation diversity than no or low costs of burning (Figure S25). This is likely related to the fact that a burning cost reduces the amount of burning over the course of model run, allowing for a landscape that is not dominated by burnt patches.

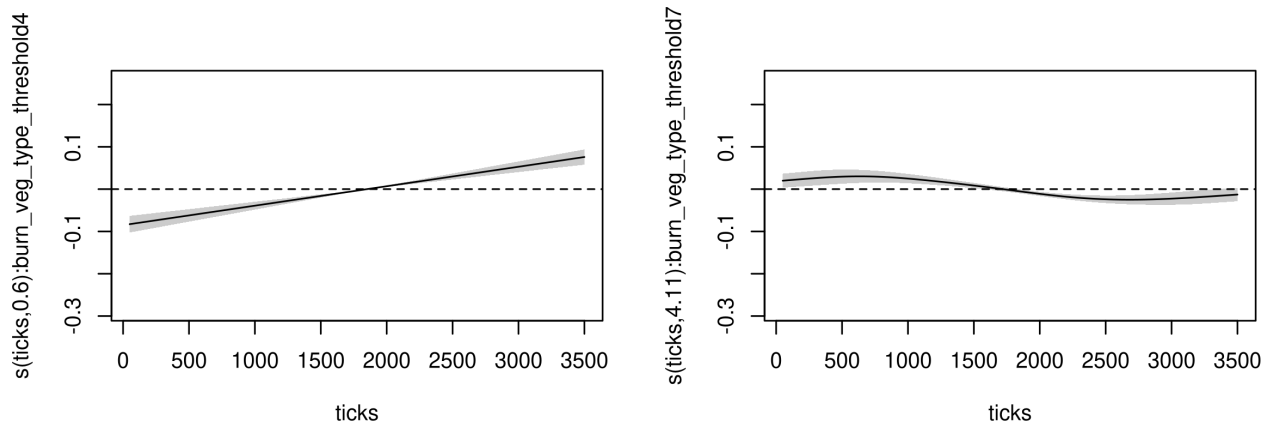


Figure S26: Effect through time of limits on vegetation types that can be burned on vegetation diversity

Similarly, stricter limits on what types of vegetation can be burned also allows for more vegetation diversity by the end of model run (Figure S26).

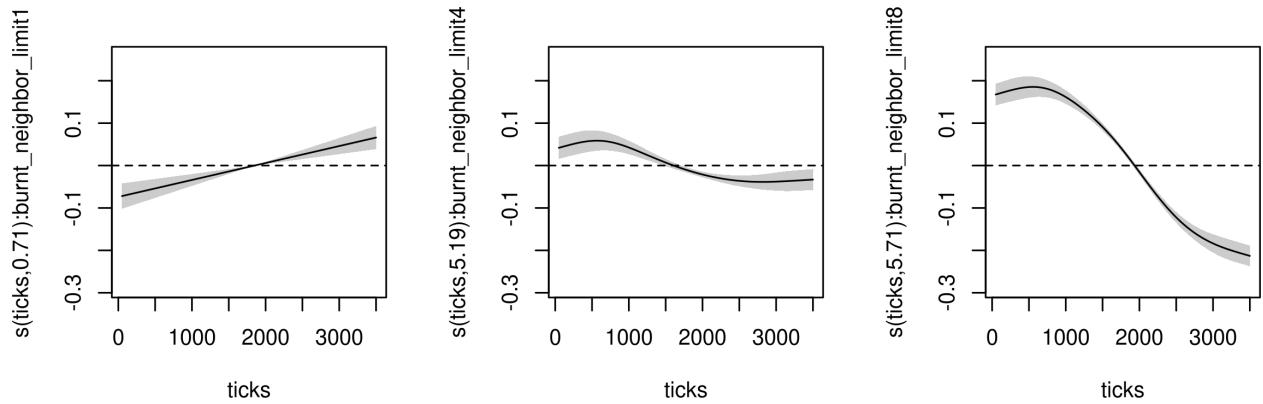


Figure S27: Effect through time of limits on burning with adjacent burnt neighbors on vegetation diversity

This same pattern is seen even more distinctly with the stricter burnt neighbor limits. When agents can burn patches with 8 burnt neighbors, vegetation diversity declines much more by the end of model run as the number of burnt patches increases (Figure S27).

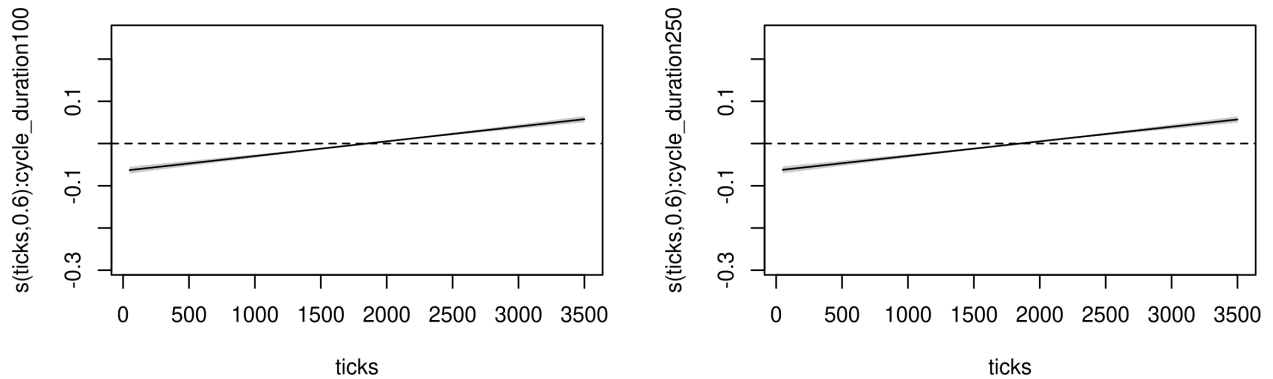


Figure S28: Effect through time of climatic cycle duration on vegetation diversity

Both values of cycle duration have similar effects (Figure S28).

6.2 Vegetation diversity under directed walks

We fit a second GAM to vegetation type diversity for model runs with directed walks.

```
veg.data.d = veg.data %>% filter(movement_model == "\"Directed Walk\"")
reg.params = parameters = c(
  "cycle_duration", # 100, 250
  "veg_cycle_start", # productive, unproductive
  "veg_distribution", # random, clustered
  "burnt_neighbor_limit", # 1, 4, 8
  "burn_cost", # 0, 50
  "burn_veg_type_threshold"
)
veg.data.d[,reg.params] = lapply(veg.data.d[,reg.params], as.factor)

dw.gam1 = gam(shannon.div ~
  #s(ticks, by = natural_ignition) +
  s(ticks, by = veg_cycle_start) +
```

```

s(ticks, by = veg_distribution) +
s(ticks, by = burn_cost) +
s(ticks, by = burn_veg_type_threshold) +
s(ticks, by = as.factor(burnt_neighbor_limit)) +
s(ticks, by = cycle_duration), data = veg.data.d)

dw.gam2 = gam(shannon.div ~ veg_cycle_start +
veg_distribution +
burn_cost +
burn_veg_type_threshold +
burnt_neighbor_limit +
cycle_duration +
#s(ticks, by = natural_ignition) +
s(ticks, by = veg_cycle_start) +
s(ticks, by = veg_distribution) +
s(ticks, by = burn_cost) +
s(ticks, by = burn_veg_type_threshold) +
s(ticks, by = burnt_neighbor_limit) +
s(ticks, by = cycle_duration), data = veg.data.d)

anova(dw.gam1, dw.gam2, test = "Chisq")

## Analysis of Deviance Table
##
## Model 1: shannon.div ~ s(ticks, by = veg_cycle_start) + s(ticks, by = veg_distribution) +
## s(ticks, by = burn_cost) + s(ticks, by = burn_veg_type_threshold) +
## s(ticks, by = as.factor(burnt_neighbor_limit)) + s(ticks,
## by = cycle_duration)
## Model 2: shannon.div ~ veg_cycle_start + veg_distribution + burn_cost +
## burn_veg_type_threshold + burnt_neighbor_limit + cycle_duration +
## s(ticks, by = veg_cycle_start) + s(ticks, by = veg_distribution) +
## s(ticks, by = burn_cost) + s(ticks, by = burn_veg_type_threshold) +
## s(ticks, by = burnt_neighbor_limit) + s(ticks, by = cycle_duration)
## Resid. Df Resid. Dev      Df Deviance  Pr(>Chi)
## 1      64261      1042.36
## 2      64251       759.73 10.647    282.63 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

dw.fit1 = lm(shannon.div ~ ticks*(.), data = veg.data.d %>%
select_at(c("ticks", "shannon.div", reg.params)))

AIC(dw.gam2)

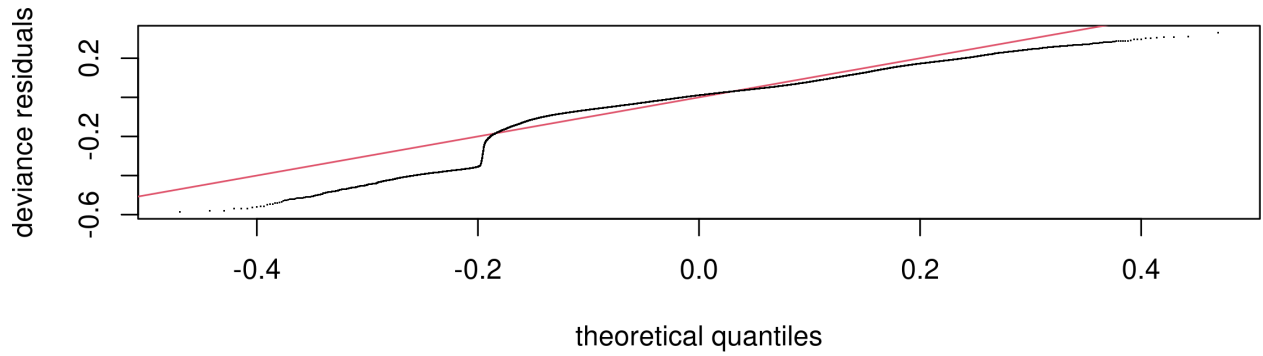
## [1] -102835.1

AIC(dw.fit1)

## [1] -97093.22

qq.gam(dw.gam2)

```



The QQ plot shows okay model performance, but for comparison to random walk model scenarios we continued with a GAM.

Overall with directed walks, we get very similar parameter effects through time for most of the parameters. There are, however, key differences in the effects of burning costs and limits to where burning can occur.

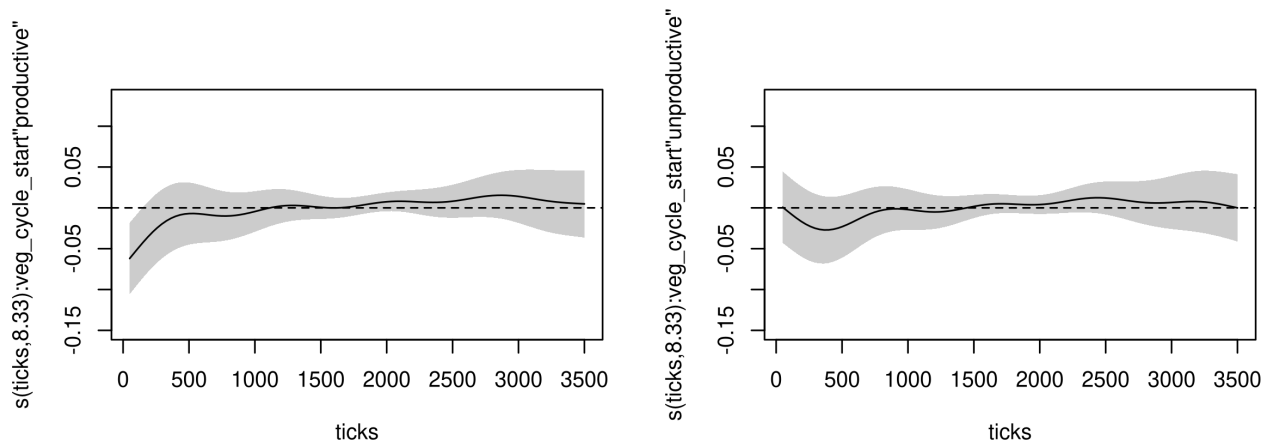


Figure S29: Effect through time of type of starting climatic period on vegetation diversity

We again see similar effects of both starting vegetation conditions on vegetation diversity across model run (Figure S29).

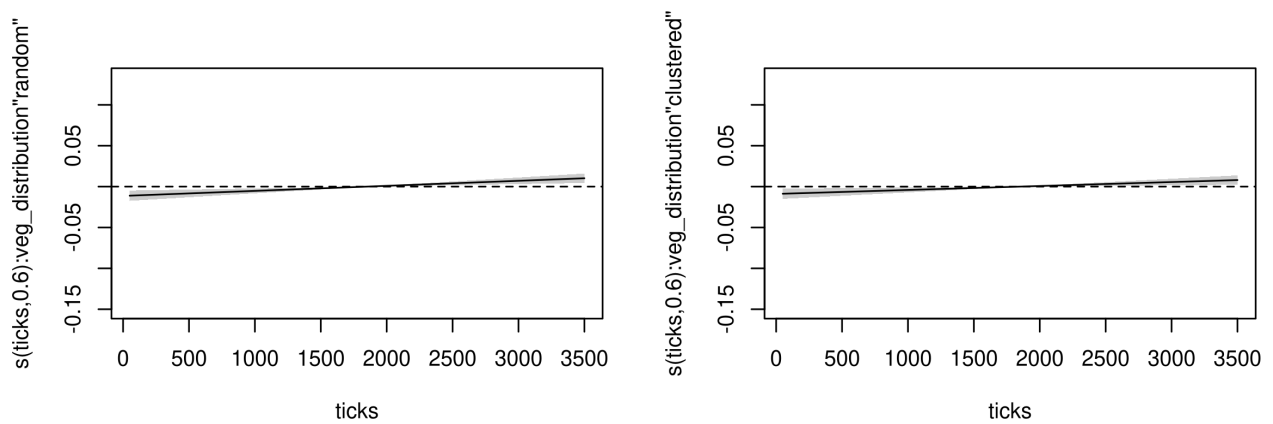


Figure S30: Effect through time of underlying distribution of vegetation types on vegetation diversity

The underlying distribution of resource potential, or maximum vegetation type value, have essentially no

effect on vegetation diversity with directed walks (Figure S30). This means that the human behavior is more important for structuring the environment with this movement model.

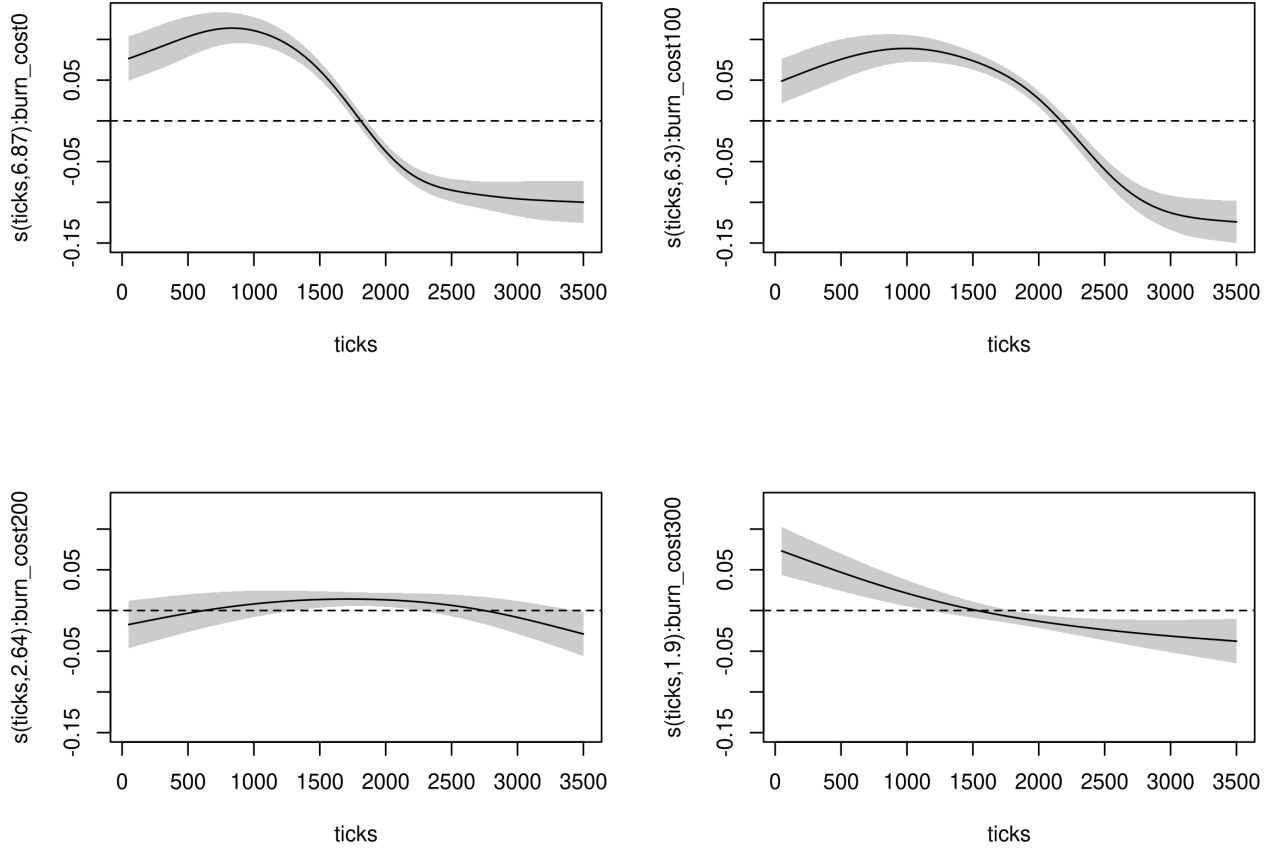


Figure S31: Effect through time of burn cost on vegetation diversity

With directed walks, we can see that low and medium costs for burning behaviors have similar effects (Figure S31): at first negative, since burning is required to introduce new vegetation types, but then becoming positive as more burning is allowed to occur. We see a similar initial pattern with no burning costs; however, by the end of model run, the huge amount of burning leads to too many burnt patches that decreases the overall diversity of vegetation on the landscape. Higher burning costs has the similar effect patterns, but a much more limited effect size. This is likely because high burning costs severely limits burning behaviors and therefore does not have a substantial impact on what vegetation types are available on the landscape.

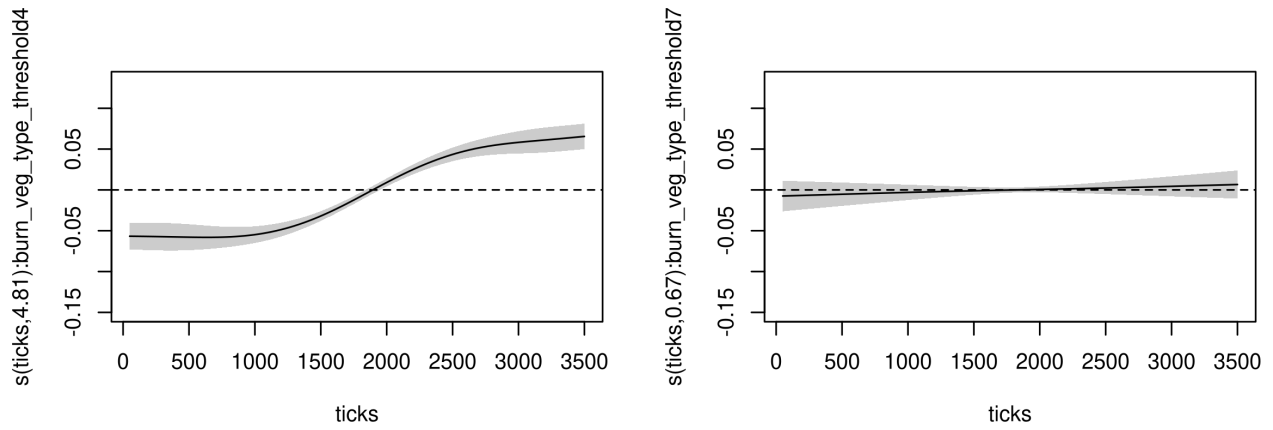


Figure S32: Effect through time of limits on vegetation types that can be burned on vegetation diversity

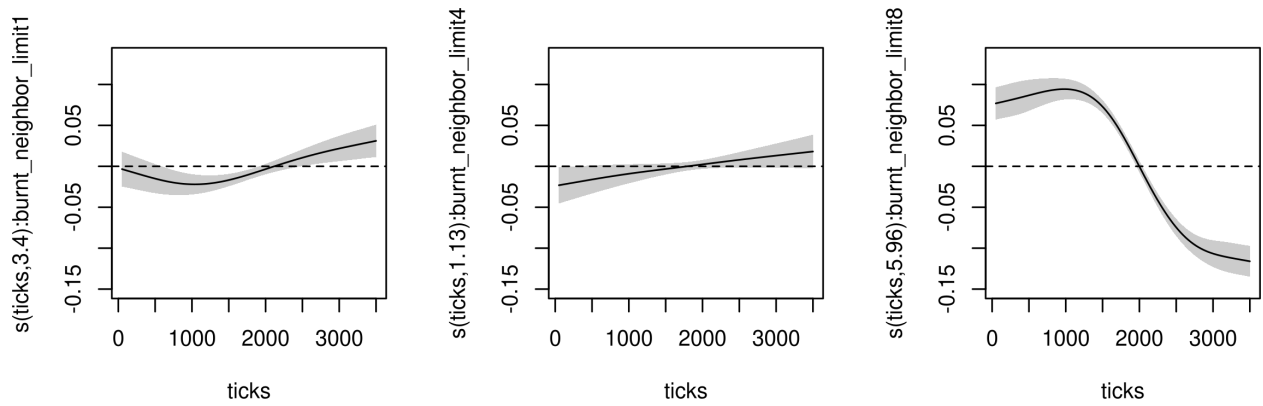


Figure S33: Effect through time of limits on burning with adjacent burnt neighbors on vegetation diversity

The limits on burning behaviors have more exaggerated effects with directed walks (Figures S32 & S33). We can again see clearly that more burning (with relaxed burning restrictions) reduces overall vegetation diversity by the end of model run.

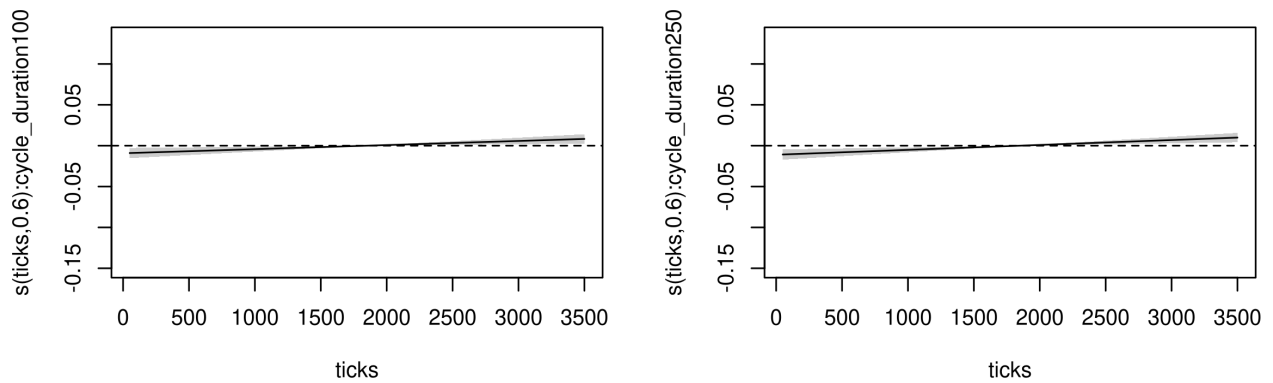


Figure S34: Effect through time of climatic cycle duration on vegetation diversity

Cycle duration appears to have no effect on vegetation diversity with directed walks (Figure S34).

7 Burning behaviors and resource distributions

The above results demonstrate that vegetation spatial autocorrelation and vegetation diversity primarily controlled by the restrictions and cost of burning behaviors; these parameters also have significant impacts on the evolution of burning behaviors across model run.

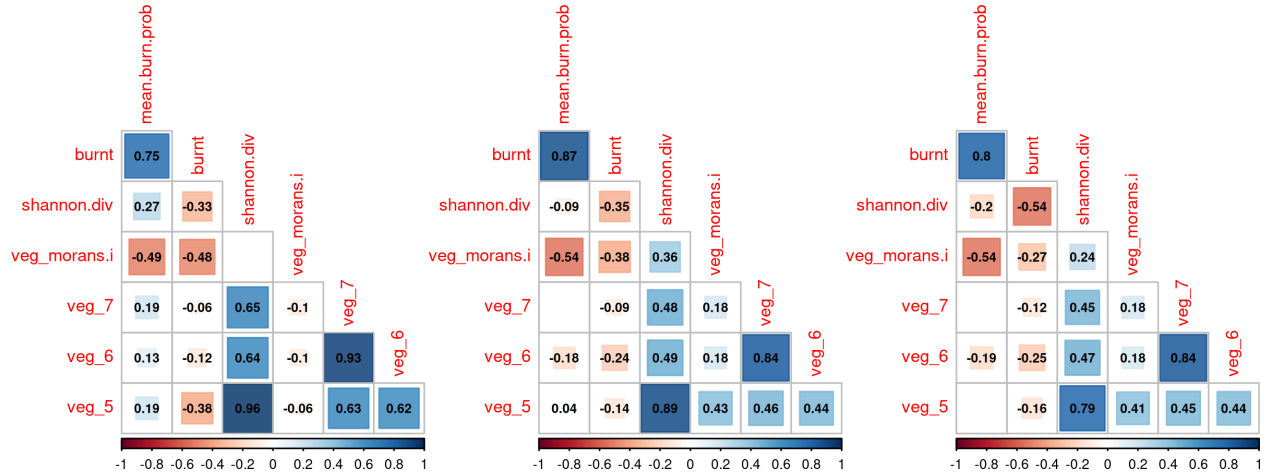


Figure S35: Correlations between burning probabilities, burnt landscape proportions, vegetation diversity, vegetation spatial autocorrelation, and high value vegetation types at beginning, middle, and end of model run from left to right

By the end of model run we see a strong negative correlation between vegetation Moran's I and Shannon Diversity (Figure S35). Increasing burning results in less diversity because of dominating proportion of burnt patches on the landscape when burning is very frequent. The negative correlation with vegetation spatial autocorrelation demonstrates that the landscape becomes more locally heterogeneous as burning becomes more frequent. Interestingly, however, there does not appear to be a significant correlation between average burning probabilities and the number of high yield patches on the landscape.

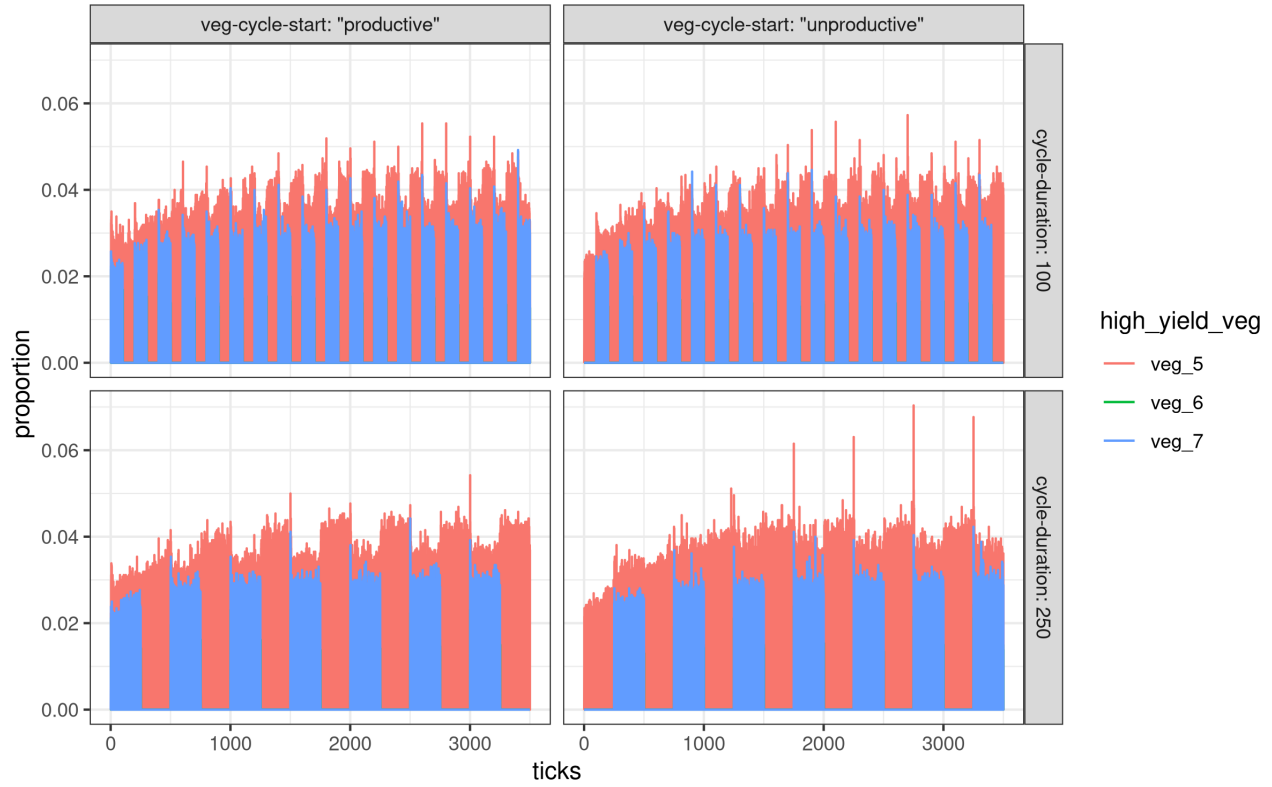


Figure S36: Proportions of high yield vegetation proportions throughout model run

However, we can see that the proportion of high yield patches increases over the course of model run (Figure S36). This means that as burning behaviors become more frequent with time, there will be more high yield patches on the landscape.

8 Forager mobility

In the model, foragers can adjust how many steps they take during a foraging bout if they determine they will meet their energy requirements with fewer steps. Movement in the model is, therefore, patterned primarily by the energy available in the gridded landscape, which is determined by whether the model is in a productive or unproductive climatic period and how long those climatic periods last.

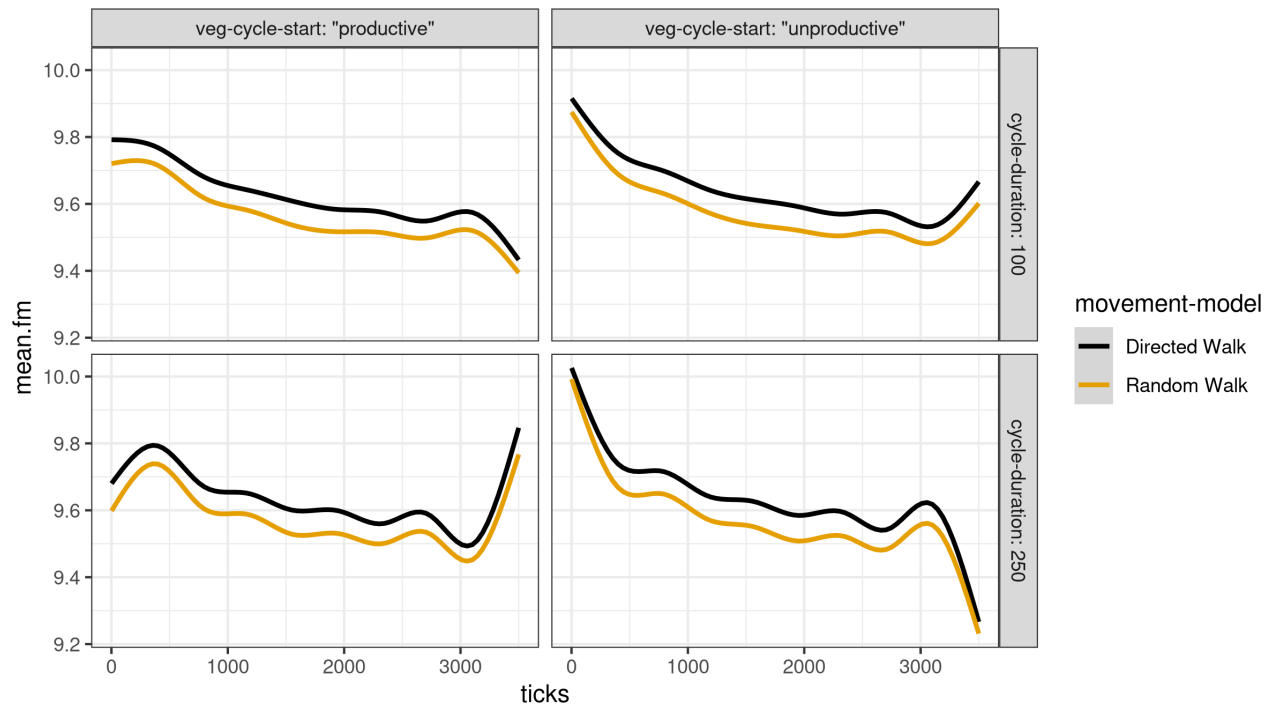


Figure S37: Trends in average steps per foraging bout across model run

We can see in Figure S37 that there is a decrease in movement during productive periods and an increase in movement during unproductive periods of the model. This is more smoothed when cycle duration is shorter.

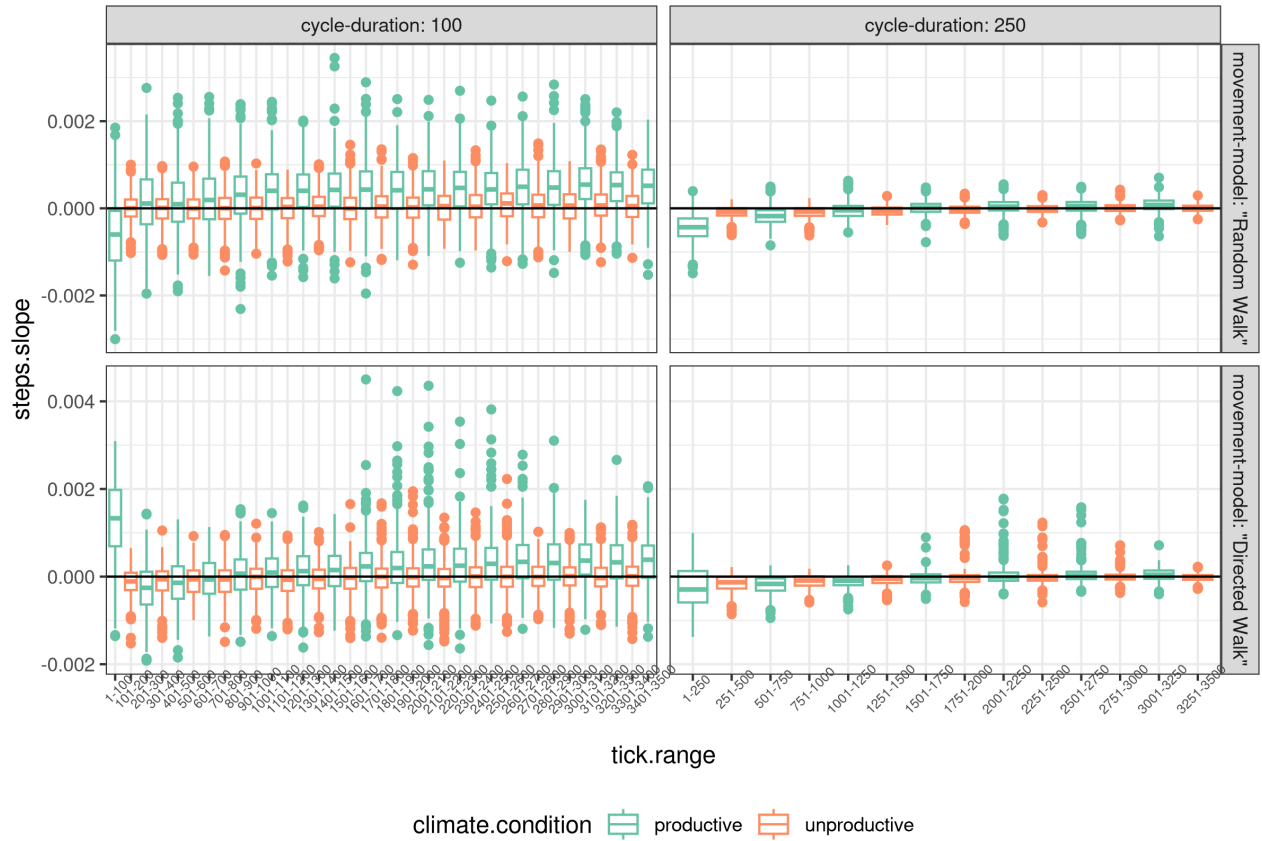


Figure S38: Slope of linear relationship between steps per foraging bout and time for each climatic interval across model run

When we break this down by climatic interval, we can see that most of the change to mobility occurs during productive intervals (Figure S38). Interestingly, across these productive intervals, it primarily at the beginning of model run when mobility decreases across the interval. By the end of model run, there is actually a slight increase in mobility across the climatic interval.

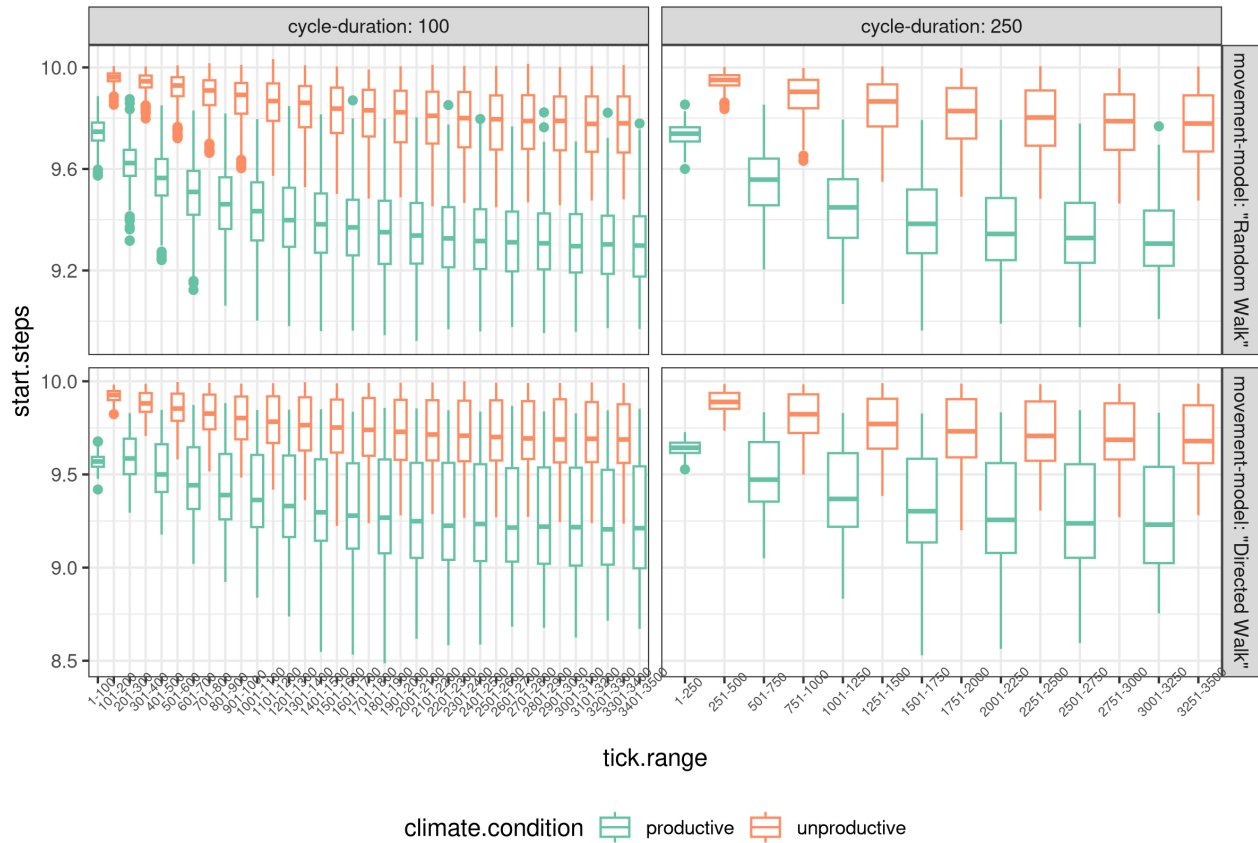


Figure S39: Average number of steps per foraging bout at the beginning of each climatic interval across model run

The overall decrease in mobility across model run is primarily caused by steep dips in mobility at the beginning of each productive interval (Figure S39).

8.1 Effect of model parameters on forager mobility

To understand how model parameters affected forager mobility, we first chose to investigate average steps per foraging bout as a linear trend to understand overall evolution during model run. Using mixed linear models, we can see how the other parameters affect forager movement while controlling for starting vegetation state to deal with the model end effects seen above.

```
mm.fit1.1 = lmer(mean.fm ~ ticks + (1 | `veg-cycle-start`), data = data)
```

```
mm.fit1.2 = lmer(mean.fm ~ ticks +
  `movement-model` +
  `burnt-neighbor-limit` +
  `burn-cost` +
  `burn-veg-type-threshold` +
  `cycle-duration` +
  `veg-distribution` +
  (1 | `veg-cycle-start`),
  data = data)
```

```
mm.fit1.3 = lmer(mean.fm ~ ticks*(`movement-model` +
  `burnt-neighbor-limit` +
```

```

        `burn-cost` +
        `burn-veg-type-threshold` +
        `cycle-duration` +
        `veg-distribution`) +
    (1 | `veg-cycle-start`),
    data = data)

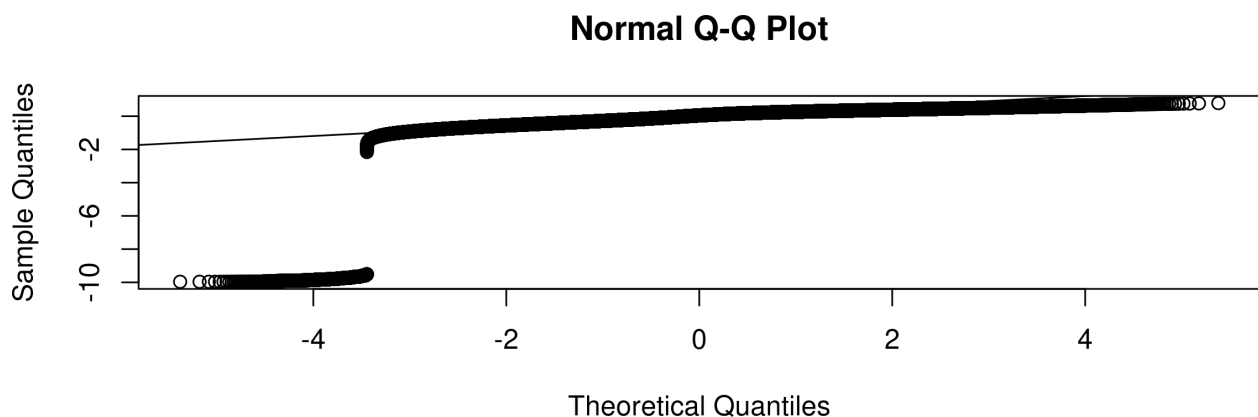
mm.fit1.4 = lmer(mean.fm ~ ticks*(`movement-model` +
    `burnt-neighbor-limit` +
    `burn-cost` +
    `burn-veg-type-threshold` +
    `veg-distribution`) +
    (1 | `veg-cycle-start`) +
    (1 | `cycle-duration`),
    data = data)

anova(mm.fit1.1, mm.fit1.2, mm.fit1.3, mm.fit1.4)

## Data: data
## Models:
## mm.fit1.1: mean.fm ~ ticks + (1 | `veg-cycle-start`)
## mm.fit1.2: mean.fm ~ ticks + `movement-model` + `burnt-neighbor-limit` + `burn-cost` + `burn-veg-type`
## mm.fit1.4: mean.fm ~ ticks * (`movement-model` + `burnt-neighbor-limit` + `burn-cost` + `burn-veg-type`
## mm.fit1.3: mean.fm ~ ticks * (`movement-model` + `burnt-neighbor-limit` + `burn-cost` + `burn-veg-type`
##      npar      AIC      BIC    logLik deviance      Chisq Df Pr(>Chisq)
## mm.fit1.1      4 9846473 9846531 -4923233  9846465
## mm.fit1.2     13 6712930 6713118 -3356452  6712904 3133561.16  9 < 2.2e-16 ***
## mm.fit1.4     21 6496365 6496668 -3248162  6496323  216580.79  8 < 2.2e-16 ***
## mm.fit1.3     22 6496354 6496671 -3248155  6496310      13.23  1  0.0002756 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

qqnorm(resid(mm.fit1.3))
qqline(resid(mm.fit1.3))

```



The QQ plot shows relatively good model performance.

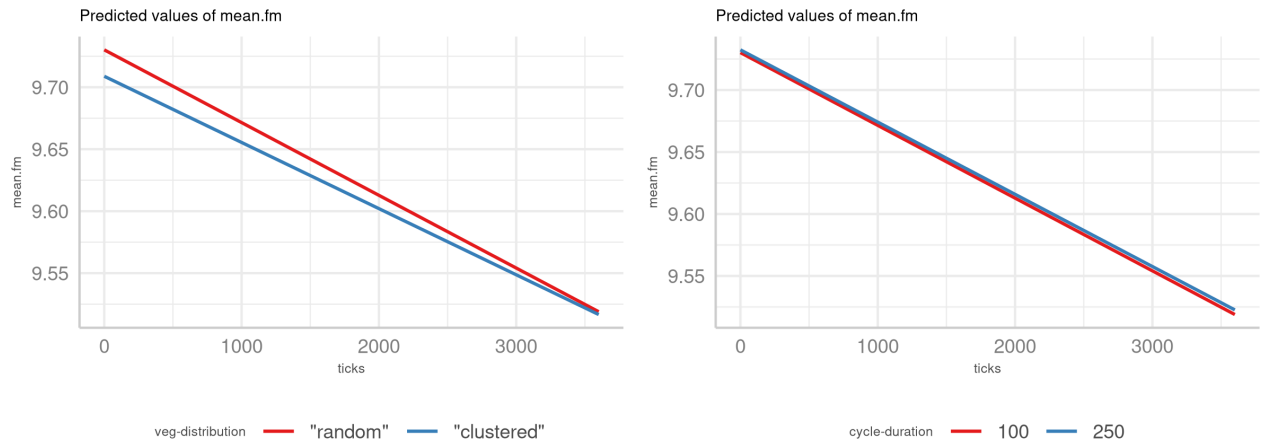


Figure S40: Predicted steps per foraging bout through time by underlying vegetation distribution and cycle duration

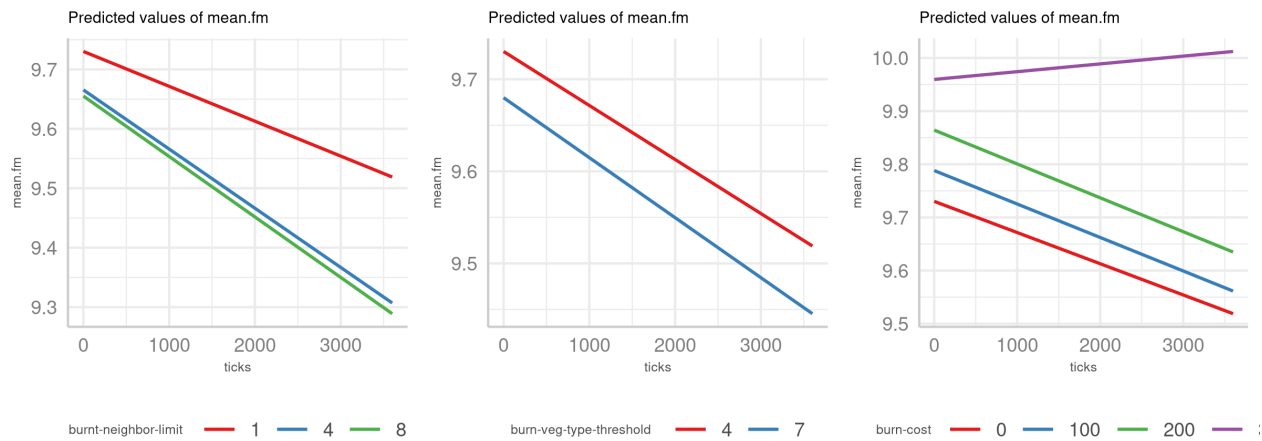


Figure S41: Predicted steps per foraging bout through time by burning costs and limitations



Figure S42: Predicted steps per foraging bout through time by movement model

The regression model demonstrate that burning restrictions and burning costs have the greatest difference in impact on forager mobility (Figure S41). Stricter limits on burning behaviors and higher costs result in

higher levels of mobility over the course of model run. Interestingly, the predicted amount of mobility for the highest burning cost shows an increase in average moves per foraging bout. This pattern is impossible in the model because we cap mobility at a maximum value. The contrast of this with the other burning costs demonstrates the differential impact that low cost niche construction strategies can have on forager behaviors.

8.2 Correlations between mobility and distributions of vegetation types

These mobility patterns due to different limits on burning behaviors occur because burning controls the energy available in the environment such that less burning means less overall energy, so agents need to move further to meet their energy needs.

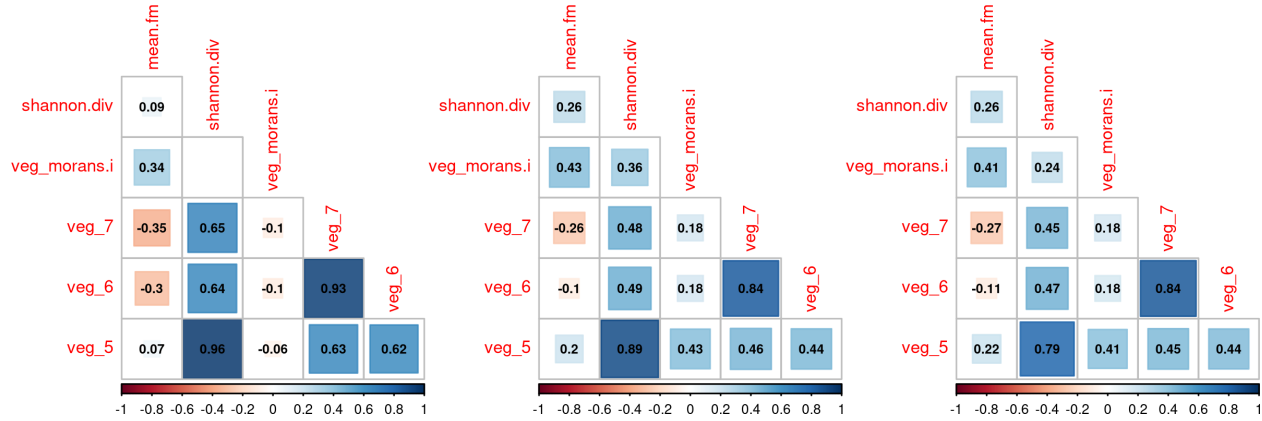


Figure S43: Correlations between forager mobility, vegetation diversity, vegetation spatial autocorrelation, and high value vegetation types at beginning, middle, and end of model run from left to right

We can see this pattern exemplified when looking at correlations between forager movement (*mean.fm*) and the highest vegetation types (*veg_7*, *veg_6*, *veg_5*), that is those that provide the most resources. We generally see negative correlations with the high vegetation types (Figure S43), indicating that low forager mobility is significantly correlated with higher proportions of these vegetation types.

We can also see that low forager mobility is highly correlated with dispersion of vegetation types, as is evidenced by the positive correlation between *mean.fm* and *veg_morans.i*. This indicates that local heterogeneity of the landscape affects how far the foragers need to move.

8.3 Correlations between forager mobility and population density

Because mobility is the behavior that agents can control in the model, we looked at how changes in this behavior changes with population evolution.

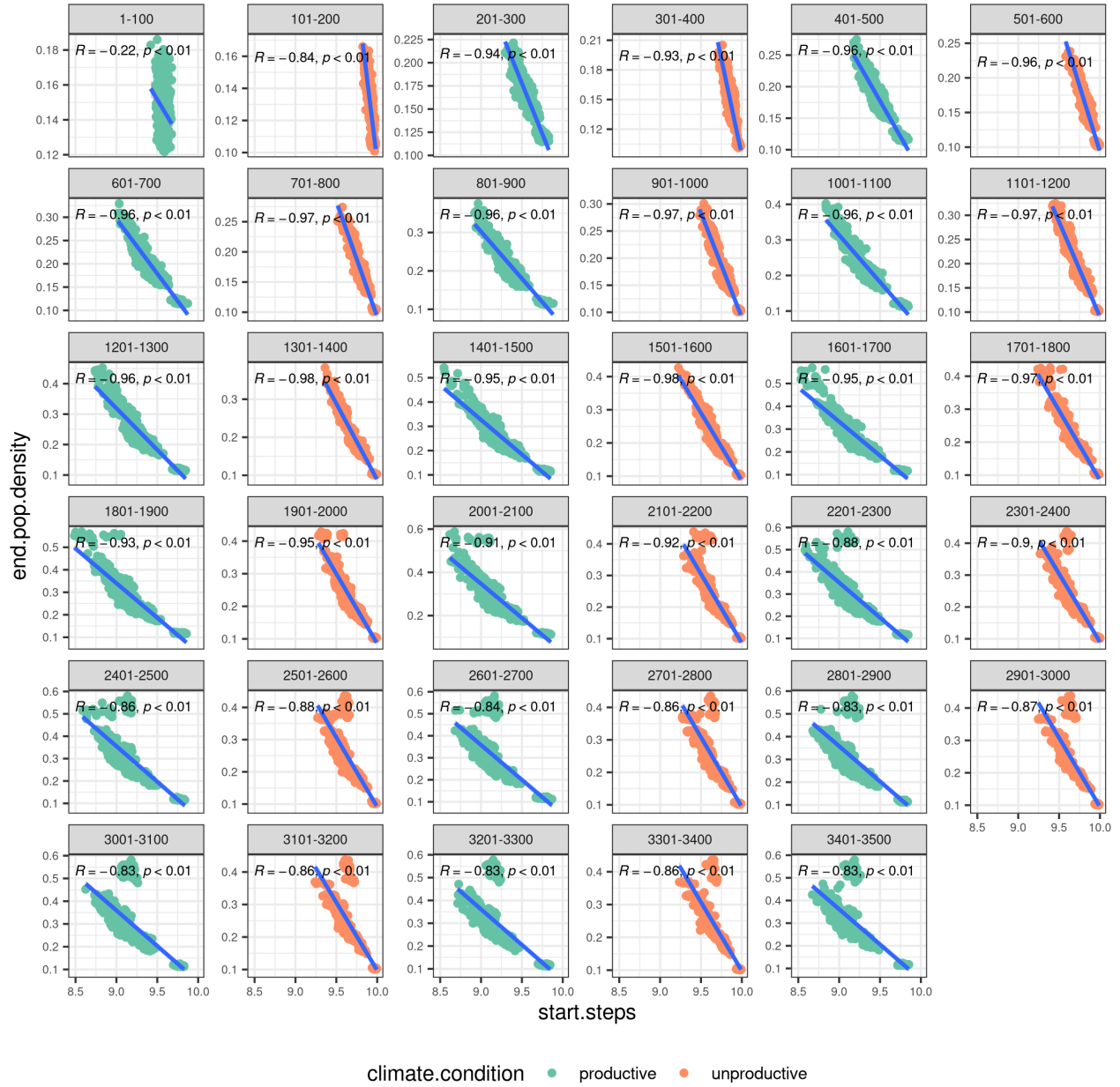


Figure S44: Relationship between average steps per foraging bout during climatic intervals and the resulting population density during the same climatic interval for model scenarios with short climatic cycles (100 ticks)

Because there is relatively little change in forager steps within climatic intervals, we looked at the relationship between the starting number of steps during a climatic interval and the resulting population density at the end of that interval. Throughout model run, these outputs are highly negatively correlated (Figure S44). This suggests that lower amounts of mobility lead to higher population densities because each foraging group requires less room to meet their foraging needs.

9 Forager interactions

When looking at forager interactions, we have to consider them in relation to populations since more foragers on the landscape will necessarily mean more possible interactions. The model outputs interaction counts

averaged across all agents and we divided this by the population count to get the average proportion of the population that an individual agent interacts with each tick.

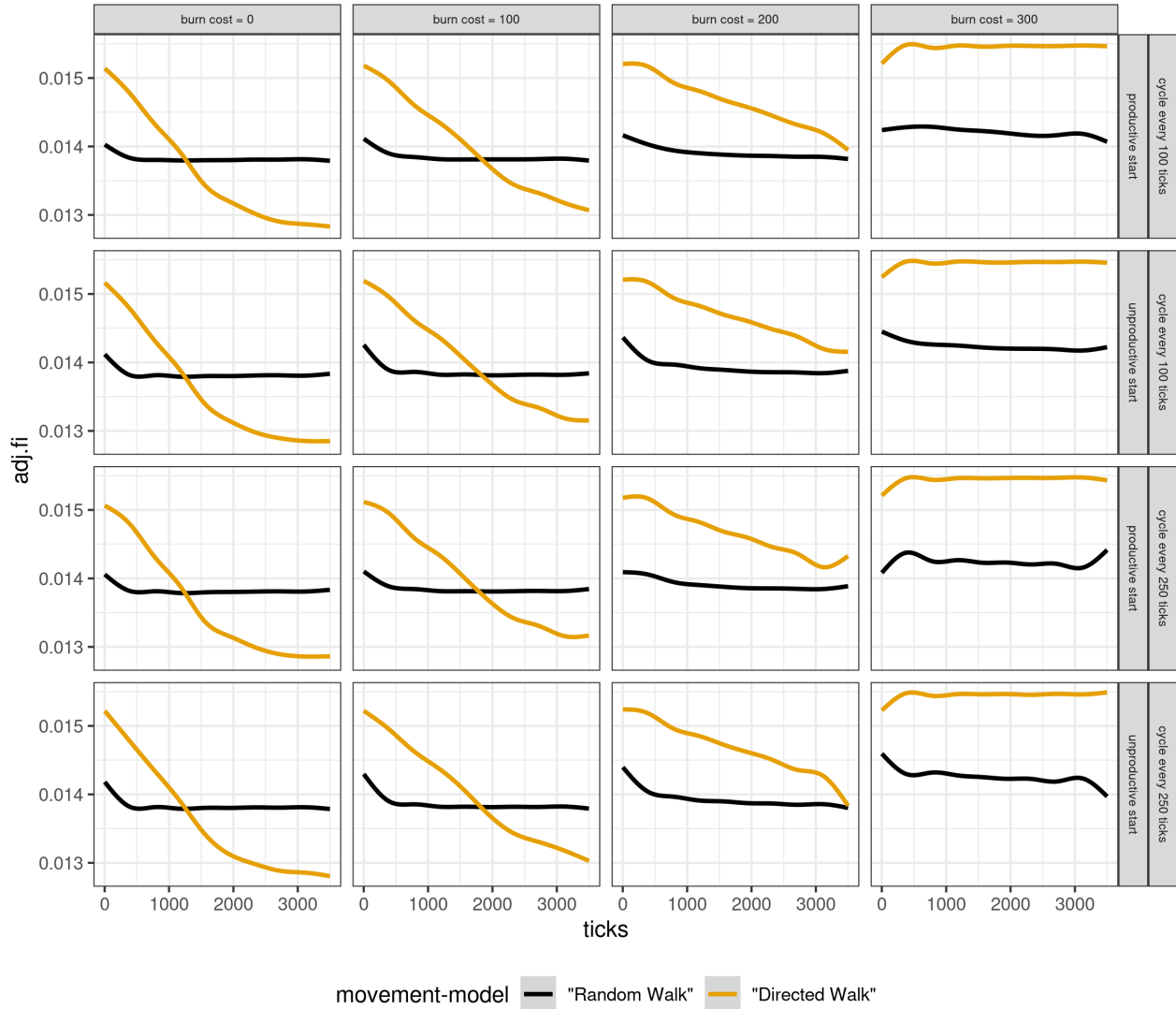


Figure S45: Trends in average proportion of population a given agent interacts with across model run

When we visualize these differences (Figure S45), we can see that for random walks, there is a steep drop off in forager interactions within approximately the first 500 ticks of the model. This is likely because populations are relatively small at the beginning of model run, so each interaction will necessarily be a greater proportion. For the subsequent analyses, we look at the patterns occurring after the first 500 ticks to understand how the parameters affect interactions after a larger population has been established.

```
fi.data = data %>% filter(ticks > 500)

#plotNormalDensity(fi.data$adj.fi)

parameters = c(
  "cycle-duration", # 100, 250
  "veg-cycle-start", # productive, unproductive
  "veg-distribution", # random, clustered
)
```

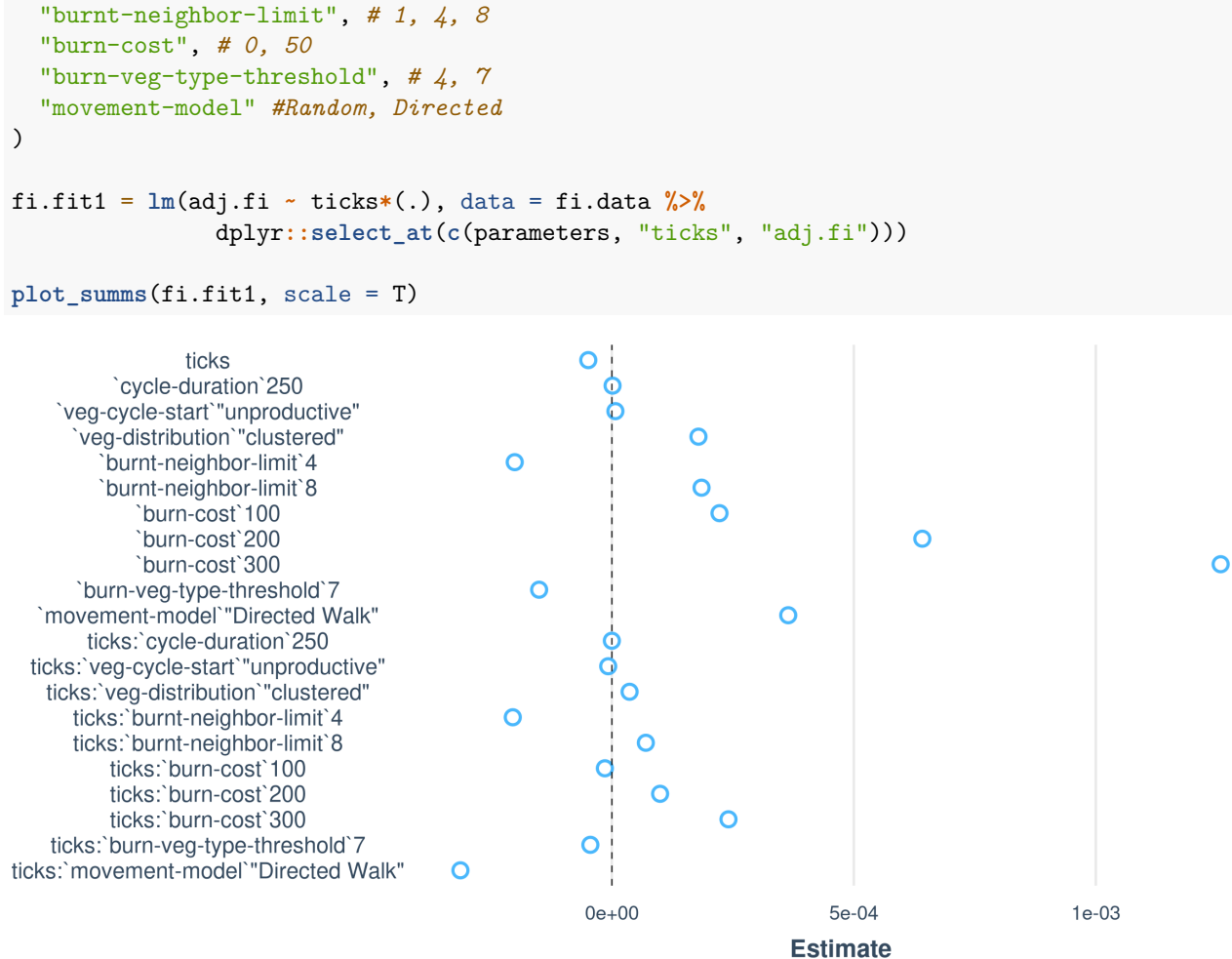


Figure S46: Standardized regression coefficients for effect of model parameters and their interaction with time on average proportion of population foragers interact with across model run

By comparing standardized regression coefficients (Figure S46), we can see the greatest effect on forager interactions is movement models and burn cost. Because the shape of the trend lines is different enough between random walks and directed walks, we split the data by movement model to further investigate parameter effects.

```

rw.data = fi.data %>% filter(`movement-model` == "\"Random Walk\"")
dw.data = fi.data %>% filter(`movement-model` == "\"Directed Walk\"")

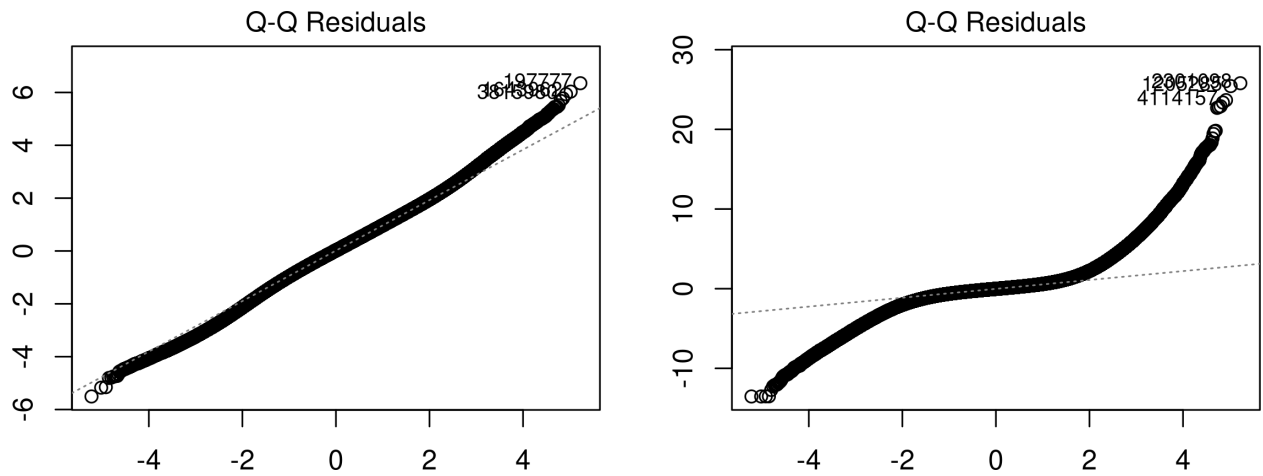
dw.fit1 = lm(adj.fi ~ ticks*(.), data = dw.data %>%
  filter(ticks > 0) %>%
  select_at(c("ticks", "adj.fi", parameters[-7])))

rw.fit1 = lm(adj.fi ~ ticks*(.), data = rw.data %>%
  filter(ticks > 0) %>%
  select_at(c("ticks", "adj.fi", parameters[-7])))

par(mfrow = c(1,2))
par(mar=c(2,2,2,2))
plot(dw.fit1, which = 2)

```

```
plot(rw.fit1, which = 2)
```



The QQ plots show relatively good model performance.

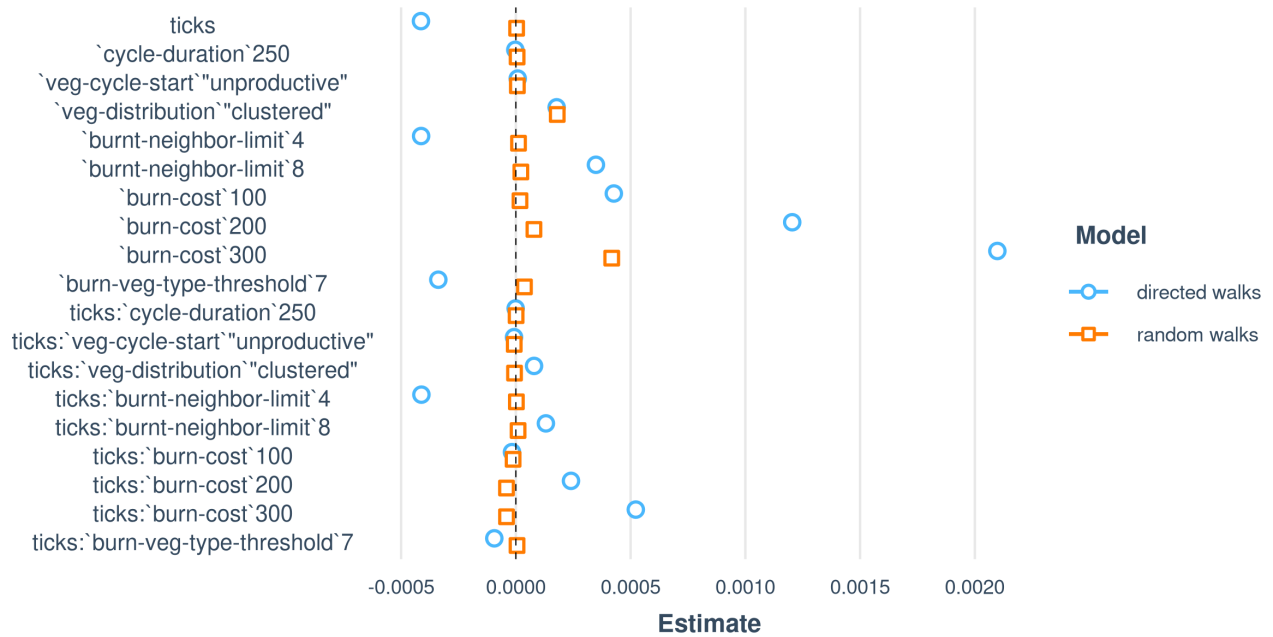


Figure S47: Standardized regression coefficients for effect of model parameters and their interaction with time on average proportion of population foragers interact with across model run split by movement model

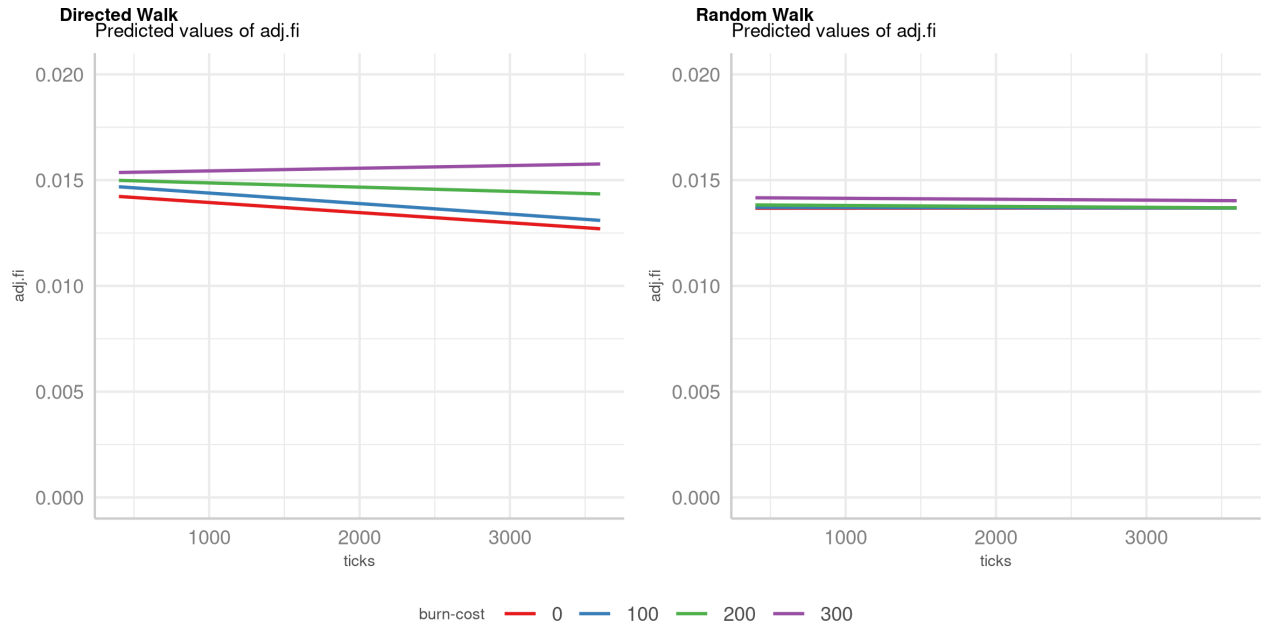


Figure S48: Predicted proportions of populations foragers interact with across model run for different burn costs and movement models

These two regressions demonstrate that restrictions/costs to burning behaviors impact forager interactions, but that these effects are much more prominent and diverging for directed walks (Figures S47 & S48). This again indicates that the amount of burning has a significant impact on the evolution of human behaviors in the model.

9.1 Change in forager interactions over time

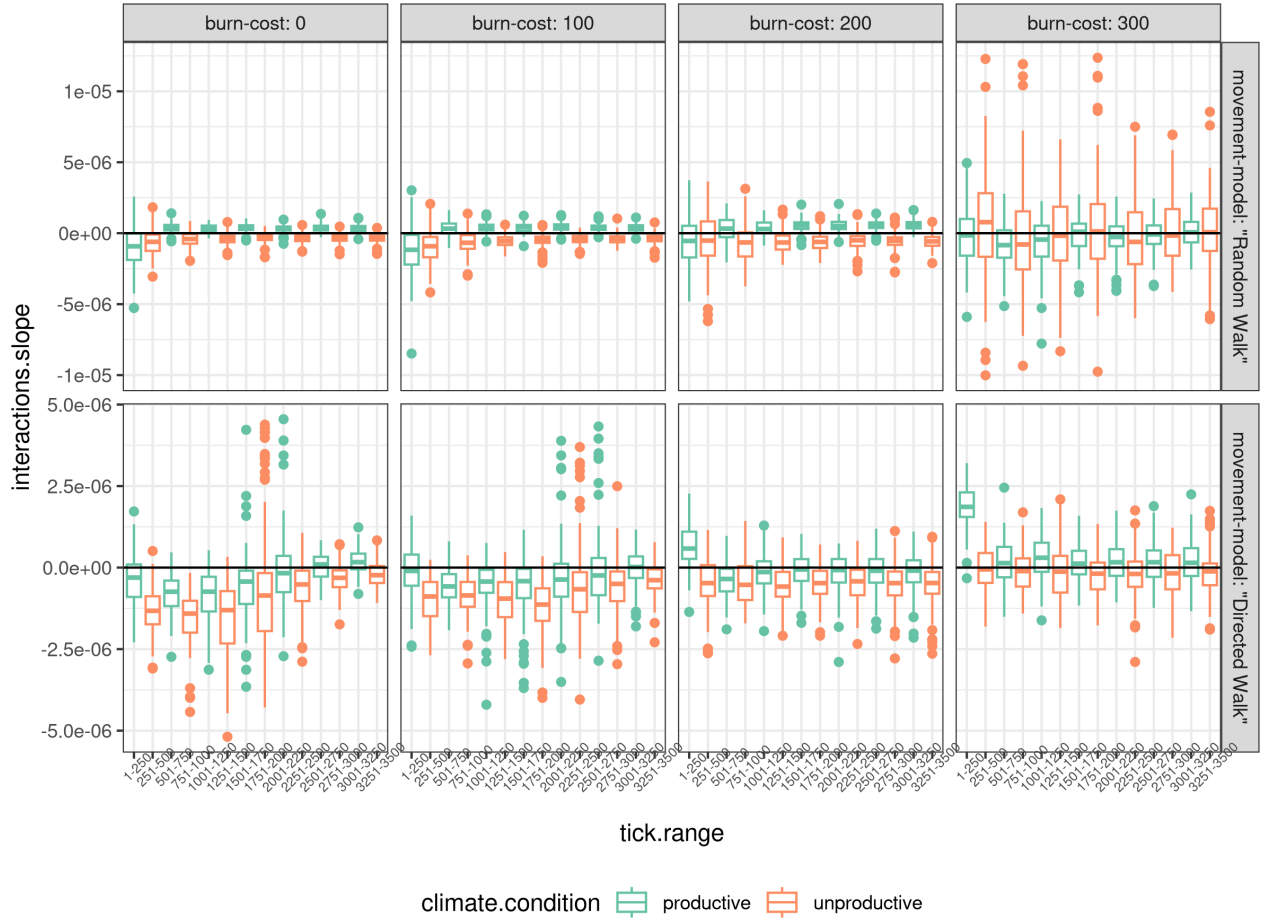


Figure S49: Change in proportion of the population foragers interact with across each climatic interval

As is expected, when we look at how forager interactions changes over across climate intervals, there is very limited change except in the earliest intervals during which populations are being established. When changes does occur, it is primarily with agents performing directed walks and when burning costs are lower (Figure S49).

9.2 Relationship between forager interactions and: forager mobility and population density

Therefore, we investigate how interaction rates over time are related to mobility and population density only with directed walks.

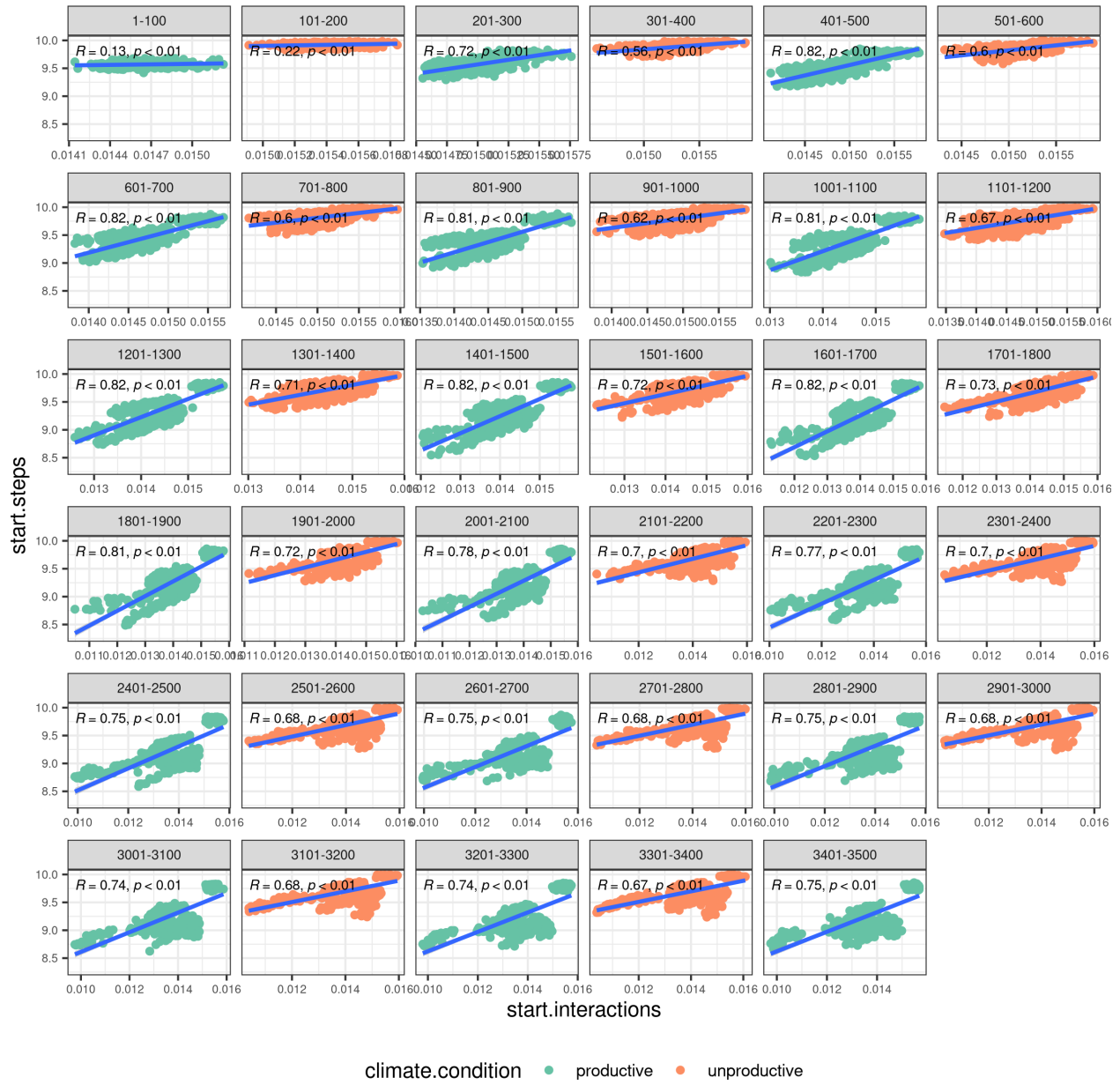


Figure S50: Relationship between proportion of population foragers interact with and steps per foraging bout at the beginning of each climatic interval across model run

Interactions are highly positively correlated with mobility throughout model run (Figure S50), meaning that more mobility allows for more interactions with directed walks.

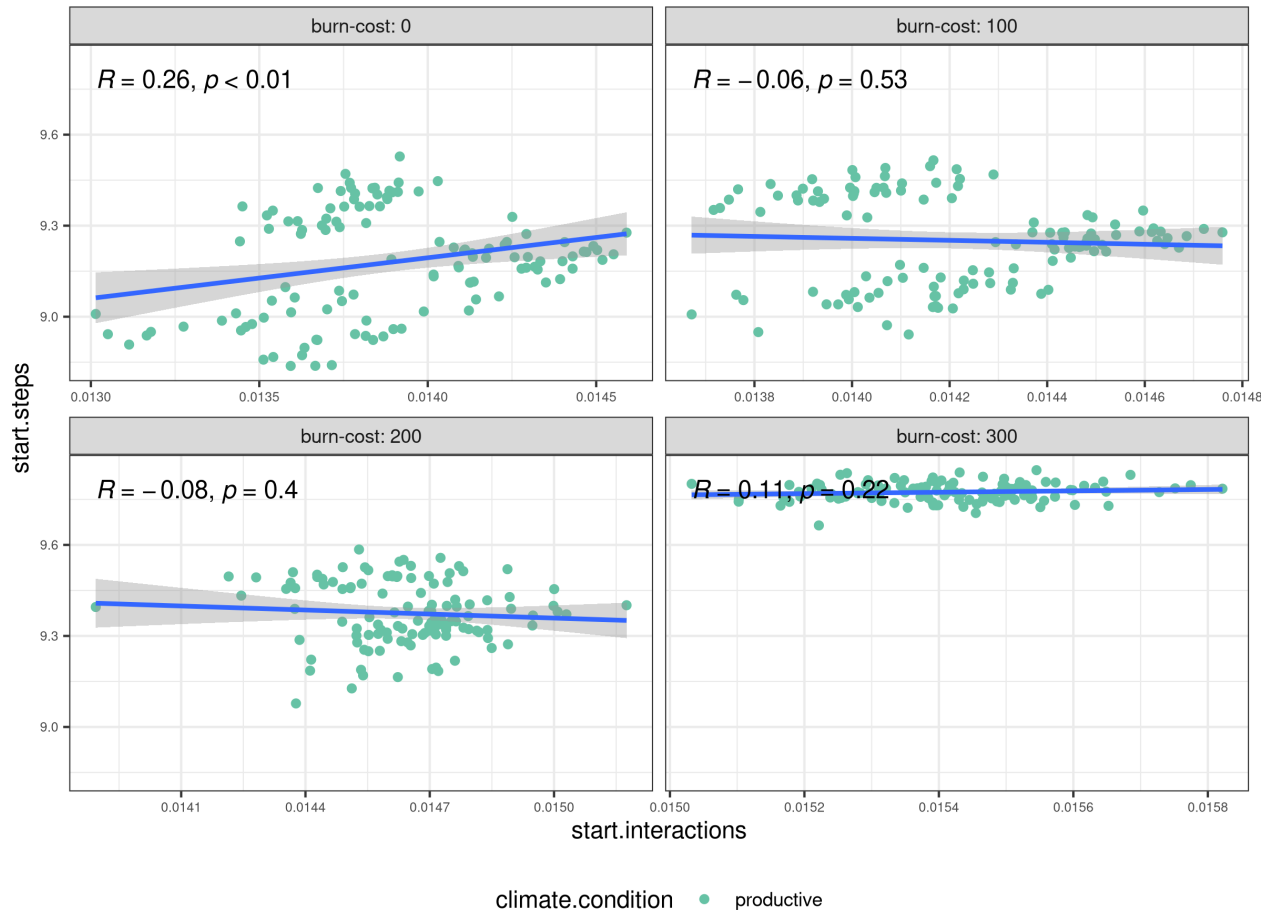


Figure S51: Relationship between proportion of the population foragers interact with and number of steps per foraging bout for a climatic period at the middle of model run. Relationship visualized by model scenarios with different burning costs

The strength of these correlations is impacted by burning costs. For example, if we take a productive climate interval in the middle of model run (Figure S51), we see significant and stronger correlations only when burning costs are low, so when burning is more frequent.

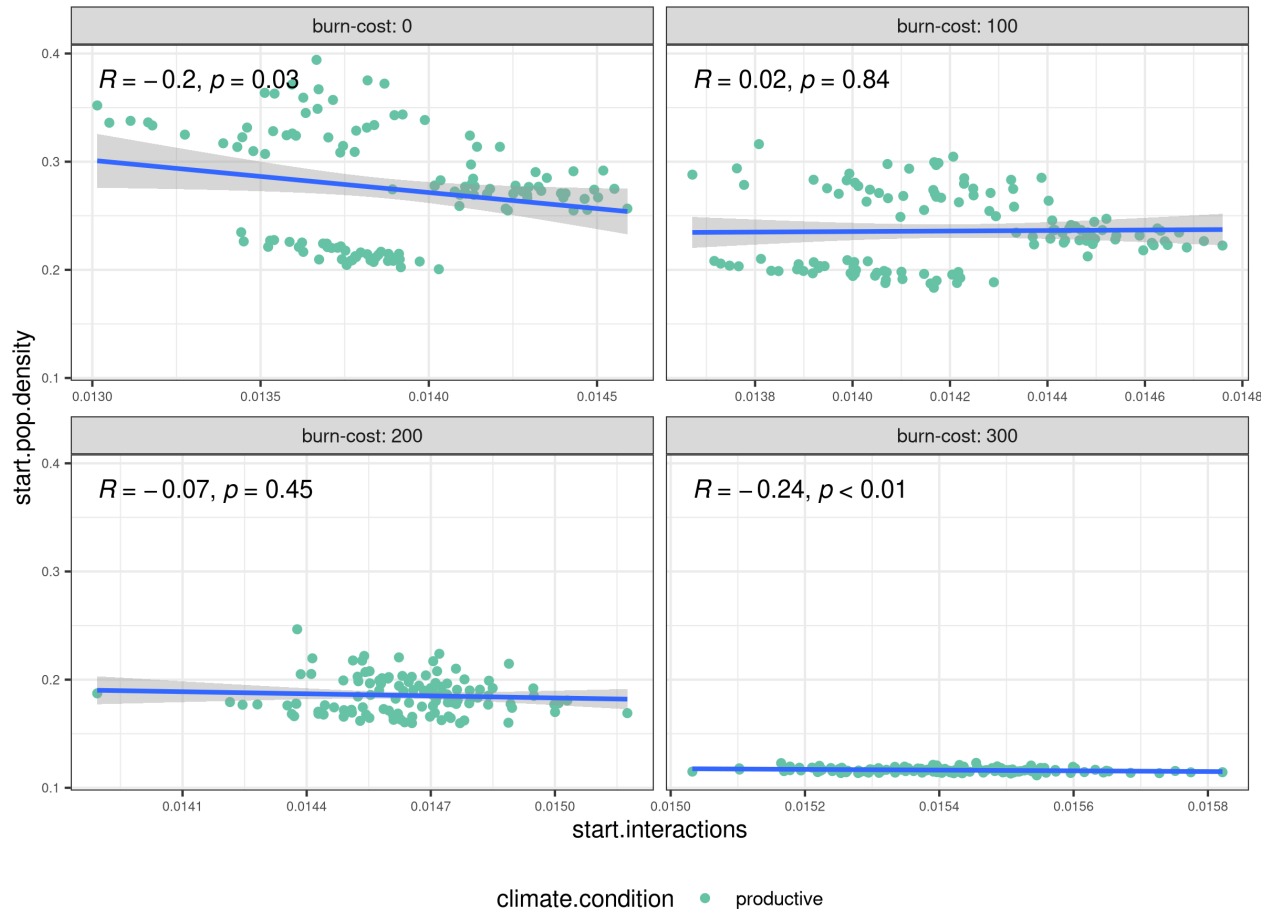


Figure S52: Relationship between proportion of the population foragers interact with and population density for a climatic period at the middle of model run. Relationship visualized by model scenarios with different burning costs

The same is true for negative correlations between interactions and population density, although here the strongest relationship is when there is no burning cost (Figure S52).

These results are because there is minimal change in forager mobility and therefore interactions at the highest burning costs.

9.3 Maintaining connections with the same foragers

In addition to understanding overall forager interaction rates, we also loosely investigated the formation of social networks. To do so, we tracked how many of a forager's interactions during a given foraging bout overlapped with the interactions of the previous bout. Doing this provides insight into whether foragers are consistently interacting with new people or if there is incipient formation of social groups. If social groups were "forming", we would expect to see increased proportions of interactions being comprised of the same foragers.

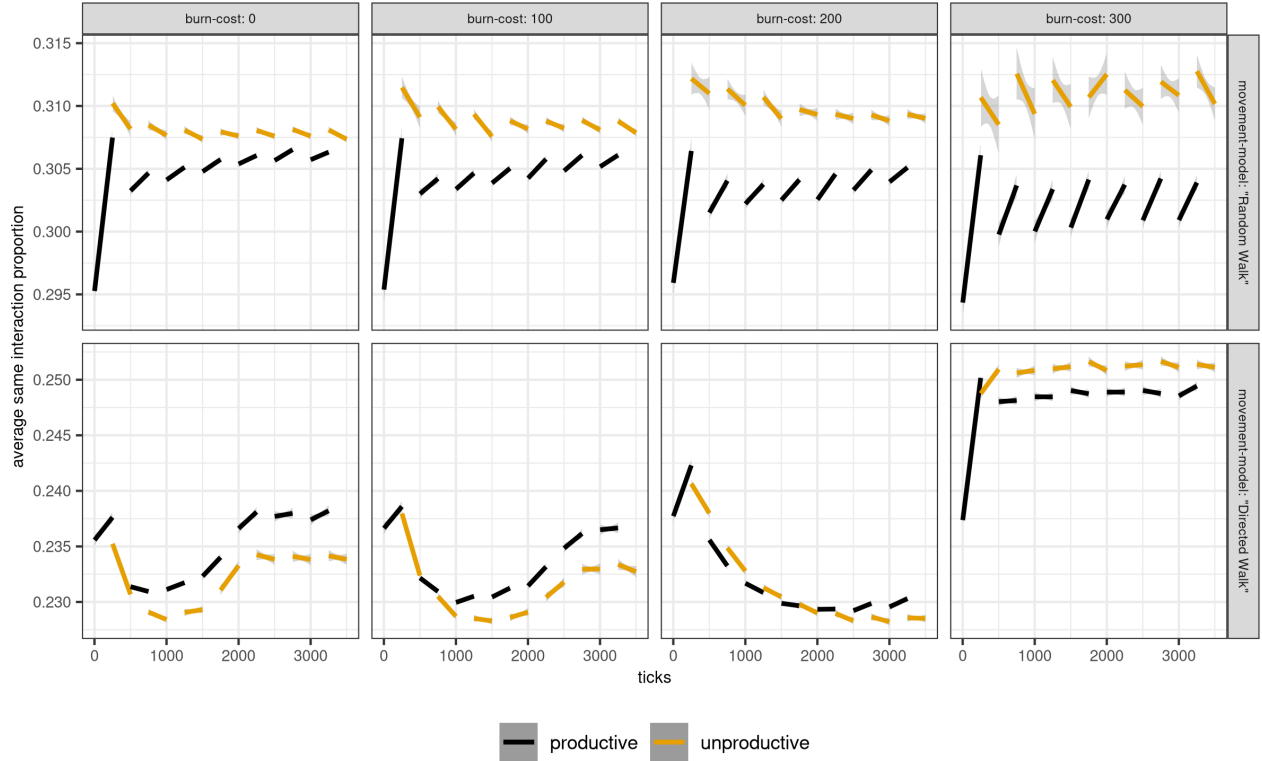


Figure S53: Trends in average proportion of interactions that are the same as the previous tick across model run. Trend lines are shown for each individual climatic interval. Results shown for model scenarios with long climatic cycles (250 ticks)

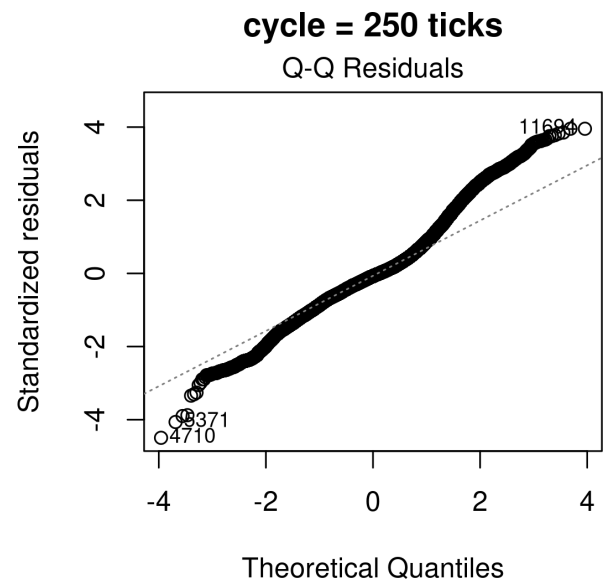
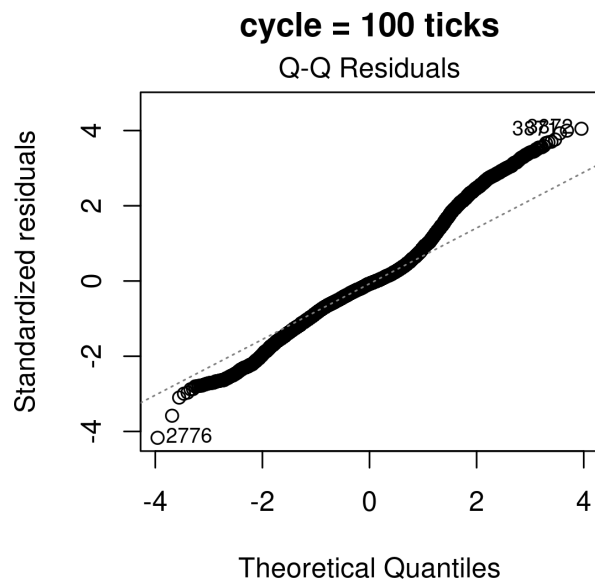
When we plot the change in average proportion of interactions that are the same as the previous tick across model run by burn cost and movement model, we see that random walks result in overall higher proportions of agents interacting with the same people compared to directed walks (Figure S53). This is driven primarily by high rates of similar interactions in unproductive periods, particularly when burning costs are high.

Conversely, with directed walks, we see an initial dip in the proportion of interactions that are with the same people as the population is establishing itself. This makes sense because the population is relatively unstable at this time, so there will be more new people on the landscape.

If we consider after that burn in period, we see that the proportion of interactions that are with the same people starts to increase again and remains higher as the population reaches equilibrium and carrying capacity (Figure S53). Because directed walks also result in fewer interactions over the course of model run, this suggests that directed walks in landscapes with a long history of burning and stable population result in smaller social networks, with foragers frequently interacting with the same people.

10 Spatial distribution of burning events

Because burning behaviors have such a significant impact on the structure of the vegetation in the environment, we were curious if this is caused by differential use of space. To examine this, we looked at the spatial autocorrelation of where burning events occur within the landscape to assess land use by the agents. This analysis is split *cycle-duration*. We first use linear regressions to assess how the parameters and time affect the spatial autocorrelation of burning events. For this, we only consider significant values of Moran's I to evaluate clustering (positive values) and dispersion (negative values).



The QQ plots show relatively good model performance.

```
fit1$coefficients[1:2] #cycle = 100 ticks
```

```
## (Intercept)      ticks
## 5.114958e-02 -3.835931e-05
```

```
fit2$coefficients[1:2] #cycle = 250 ticks
```

```
## (Intercept)      ticks
## 5.555737e-02 -3.934859e-05
```

The intercept values of the linear regressions show that, on average, burning events are on average marginally clustered on the landscape. This spatial autocorrelation is then significantly affected by time and, on average, decreases as model run progresses.

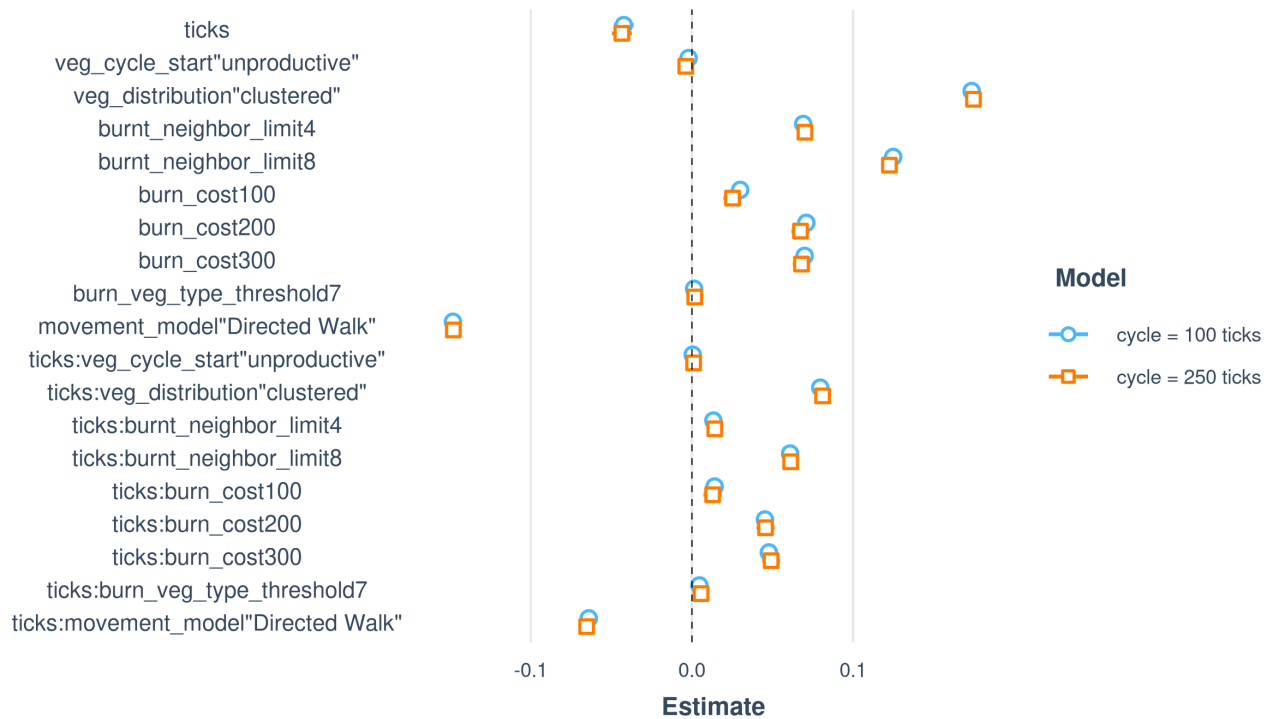


Figure S54: Standardized regression coefficients for effects of model parameters and their interactions with time on spatial autocorrelation of burning events

When looking at scaled regression coefficients, we can see that underlying distribution of resource potential (maximum veg type), burning limits, and movement models have the strongest effects on the spatial clustering of burning events (Figure S54). These strong effects are also maintained over time, as evidence by the high standardized regression coefficients for the interaction variables between these parameters and time). Importantly, time increase the effect size of these parameters in the same direction. For example, directed walks reduce the clustering of burning events on average and this effect is more negative with time.

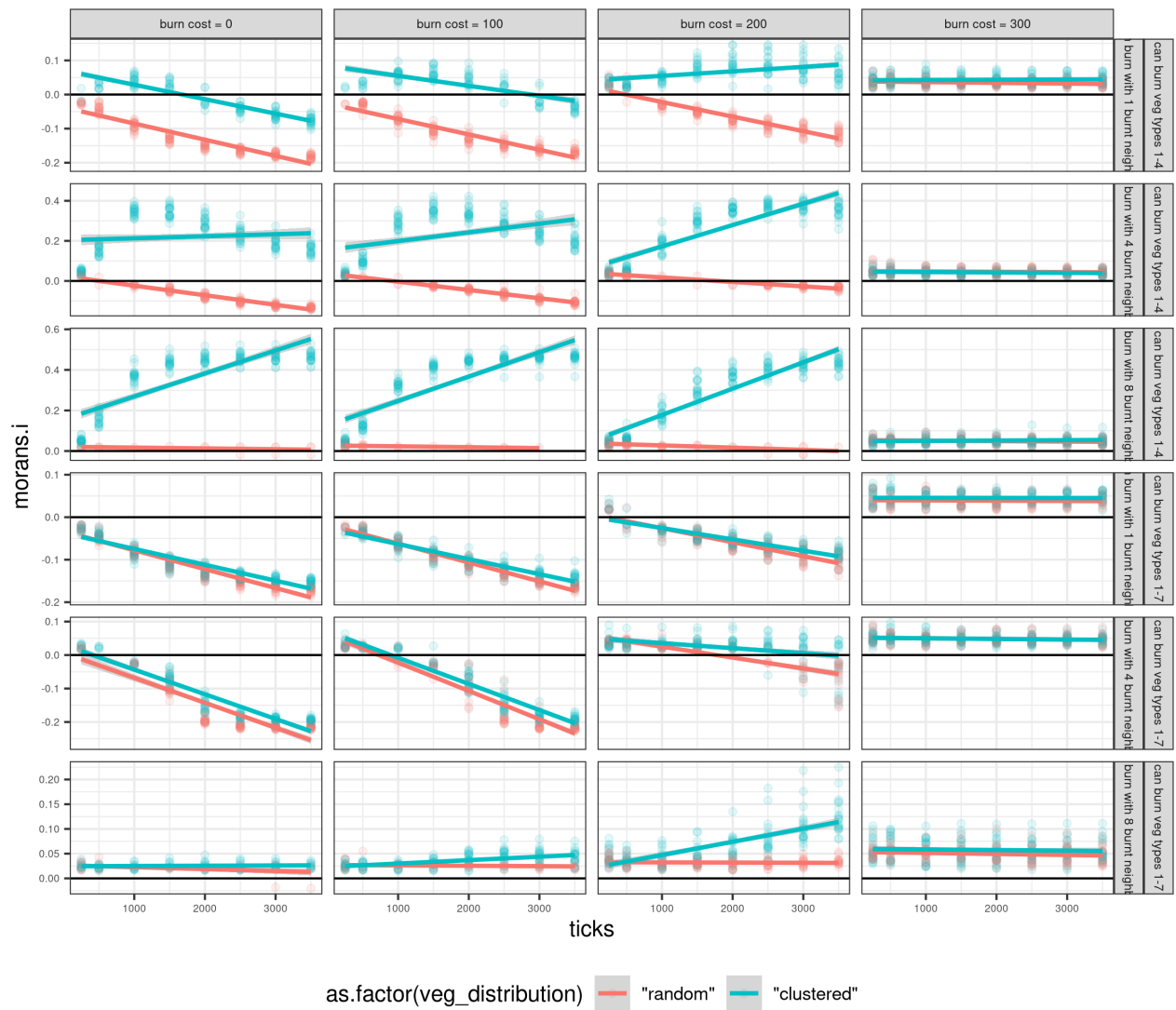


Figure S55: Trends of spatial autocorrelation of burning events across model run for model scenarios with directed walks

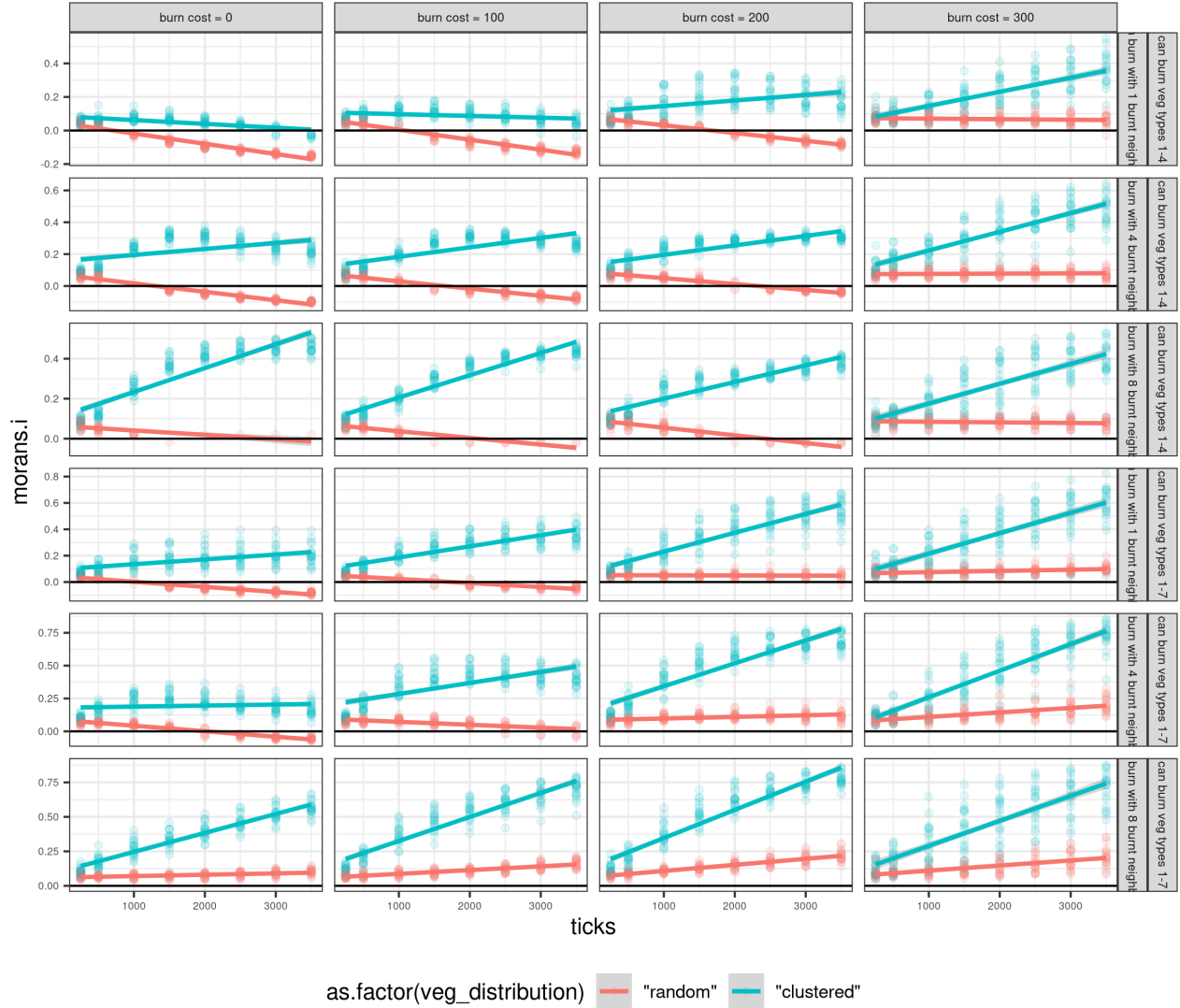


Figure S56: Trends of spatial autocorrelation of burning events across model run for model scenarios with random walks

Plotting the trends of burning event spatial autocorrelation through time by these significant parameters allows us to better visualize these patterns. Here, we split the data by movement model with trends for directed walks (Figure S55) and trend lines for random walks (Figure S56).

First, it is clear that, under random walks, underlying distribution of vegetation potential has a significant impact on clustering or dispersion of burning events (Figure S56). This is especially true when burning is limited to specific patches (can burn with fewer burnt neighbors and can only burn certain vegetation types) under low burning costs. As burning costs increase, and burning becomes more infrequent, burning events are more likely to cluster despite limits on which patches can be burnt.

With directed walks, the effect of underlying vegetation potential is more varied (Figure S55). For example, when agents can burn any vegetation type, underlying vegetation potential does not change the trend towards spatial randomness or dispersion of burning events. When agents can only burn specific vegetation types, underlying vegetation potential is more important, particularly in cases where patches can have more burnt neighbors. In these cases, underlying clustering of vegetation type potential leads to more clustering of burning events.

10.1 Relationship between location of burning events and population density

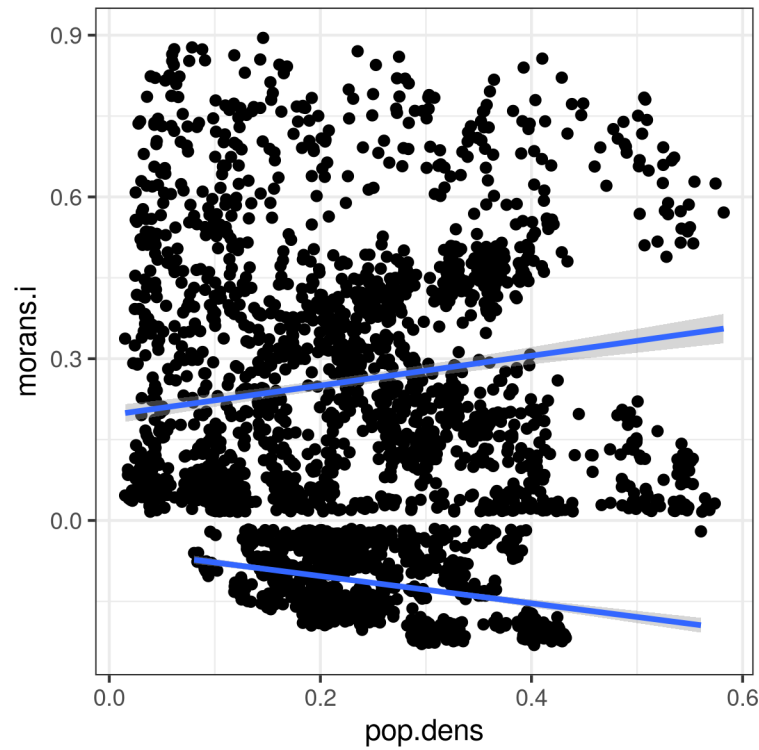


Figure S57: Relationship between population density and spatial autocorrelation of burning events at the end of model run. Linear relationships shown with blue lines for clustered burning events (positive Moran's I values) and dispersed burning events (negative Moran's I values)

In Figure S57, we can see that increased population density is clearly associated with increased dispersion of burning events at the end of model run. The correlation between clustering and population density skews positive, but the graph shows that this is less consistent at the highest population densities.

11 References

Package	Version	Citation
base	4.3.2	@base
betareg	3.1.4	@betareg2010; @betareg2012
bookdown	0.42	@bookdown2016; @bookdown2025
corrplot	0.92	@corrplot2021
fitdistrplus	1.2.1	@fitdistrplus
formatR	1.14	@formatR
ggeffects	1.7.0	@ggeffects
ggpubr	0.6.0	@ggpubr
ggsignif	0.6.4	@ggsignif
ggspatial	1.1.9	@ggspatial
ggthemes	5.1.0	@ggthemes
grateful	0.2.10	@grateful
grid	4.3.2	@grid
gtsummary	1.7.2	@gtsummary
jtools	2.2.2	@jtools
knitr	1.45	@knitr2014; @knitr2015; @knitr2023
lme4	1.1.35.1	@lme4
lmerTest	3.1.3	@lmerTest
mgcv	1.9.1	@mgcv2003; @mgcv2004; @mgcv2011; @mgcv2016; @mgcv2017
png	0.1.8	@png
psych	2.3.9	@psych
RColorBrewer	1.1.3	@RColorBrewer
rcompanion	2.4.36	@rcompanion
rmarkdown	2.25	@rmarkdown2018; @rmarkdown2020; @rmarkdown2023
scales	1.3.0	@scales
segmented	2.0.0	@segmented2003; @segmented2008; @segmented2014; @segmented2016; @segmented2017; @seg
sp	2.1.4	@sp2005; @sp2013
spdep	1.3.5	@spdep2013; @spdep2018; @spdep2022; @spdep2023
tidyverse	2.0.0	@tidyverse
vegan	2.6.6.1	@vegan